Journal of Accounting and Economics 81 (2026) 101829

Contents lists available at ScienceDirect

Journal of Accounting and Economics

journal homepage: www.journals.elsevier.com/journal-of-accounting-and-economics

Mandatory disclosure of investors' fossil fuel holdings☆

Gregory S. Miller [a,*], Douglas R. Stockbridge Jr. [b], Christopher D. Williams [a]

[a] University of Michigan, Stephen M. Ross School of Business, United States
[b] Boston College, Carroll School of Management, United States

A B S T R A C T

Regulators around the world have begun to require investment companies to provide information regarding fossil fuel investments to external stakeholders. In this paper we examine whether such disclosures impact the investment portfolios and/or investment policies of the disclosing firms. Using a 2016 California disclosure mandate that required some U.S. insurance companies to disclose their fossil fuel investments on a public website, we find the disclosing insurers reduced their fossil fuel investments by approximately 20 % relative to the non-disclosers. Despite this on-average result, we note significant variation in changes to investment portfolios. We find insurers pressured by external stakeholders, including public shareholders and environmental activists, are more likely to divest. In contrast, enhanced Californian regulatory oversight power is unrelated to divesture. Even after the disclosure mandate is reversed, we find the disclosing insurers do not revert to their pre-policy holdings of fossil fuel investments, suggesting the impact created a longer-term change in investment behavior.

1. Introduction

Recently, regulators have mandated that investment companies disclose their investments in fossil fuel securities. This requirement seeks to help capital market participants and other stakeholders better evaluate the firm's potential environmental impact. Despite the growing popularity of such mandatory environmental disclosure policies, little is known regarding how such policies impact the investment portfolios and/or investing policies of the disclosing firms. In this paper, we examine a 2016 California regulatory change that required insurance companies to publicly report their fossil fuel investments on the California Department of Insurance (CDI) website. This requirement made the information easily accessible and free for all stakeholders. Concurrently, the insurance commission explicitly called for a voluntary divestment in fossil fuel investment, suggesting the goal of disclosure was to facilitate stakeholder pressure to divest (Jones, 2016a).

Consistent with the insurance commissioner's goal of divestiture, our on-average results indicate that firms decreased their

☆ We would like to thank Ed deHaan and Michelle Hanlon (editors) as well as an anonymous referee. We received helpful comments on earlier versions of the paper from Thomas Bourveau, Paul Fischer, Luzi Hail, Mark Maffett, Catherine Schrand, Carol Seregni, Sandra Schafhäutle, Christina Zhu and workshop participants at the University of Michigan, University of Pennsylvania, and the Miami "Winter Warm-Up" Accounting Conference for helpful comments and suggestions.

* Corresponding author.
E-mail address: millerg@umich.edu (G.S. Miller).

https://doi.org/10.1016/j.jacceco.2025.101829
Received 29 May 2024; Received in revised form 29 July 2025; Accepted 14 August 2025
Available online 18 August 2025

holdings in fossil fuel companies.[1] However, there is significant variation in the response, with many firms choosing not to alter their investments or their related investment policies. Only 19 of the 314 treated firms completely divested, leaving most firms with continued investment in fossil fuel companies. Examining the within-sample decision to fully or partially divest, we find that firms with previous environmental activist lobbying or high capital market visibility are more likely to divest. Conversely, we do not find that greater exposure to California regulatory power, measured by legal incorporation or percentage of business, influences divestment decisions.

The rule we examine (hereafter the "California Rule") was novel at the time, but subsequently, policymakers in several countries have prescribed similar disclosure regulation (Duran, 2021; Halper et al., 2022; Teu, 2021; Treacy et al., 2022). However, a lack of clear disclosures and concerns about "greenwashing" has further led regulators to advocate for more explicit fossil fuel reporting (ESMA, 2022).[2] The intent of these policies is similar to the California Rule. As one regulator explained, "[the disclosure regulation] is to ensure transparency on climate action by investors, to enable the public authorities, NGOs, think tanks and, more generally, civil society to use this reporting process to put pressure on investors" (Evain et al., 2018).[3]

Despite the growing interest in California Rule-type policies, there is a lack of large-sample evidence on the disclosing firms' responses to such policies, with a few exceptions. For example, in concurrent working papers Dai et al. (2024) find a reduction in greenhouse gas emissions among European funds following the introduction of the Sustainable Finance Disclosure Regulation (SFDR), Messonier and Nguyen (2021) find disclosure leads French investors to reduce their fossil fuel holdings, and Gehricke et al. (2025) find New Zealand assets managers decrease their portfolio's carbon intensity. The absence of evidence likely stems from the recent introduction of these types of standards and research design challenges. Specifically, it is empirically challenging to estimate the effect of these policies because of the i) nonrandom selection of firms subject to the disclosure, ii) lack of a counterfactual, iii) presence of concurrent regulation, and iv) lack of data on the outcomes of interest.[4]

Our setting has several unique research design features that allow us to mitigate these challenges and provide evidence regarding the policy's efficacy. First, the disclosure applies only to insurance companies licensed in California, and due to heavy regulation, prior to the policy, both disclosing (Californian) and non-disclosing (other state) firms had similar investment portfolios, mandates, and strategies (Rabinowitz, 2019). Second, there is security-level data prior to and after the disclosure, which allows us to track portfolio changes for treated and nontreated firms.[5] Third, a concurrent insurance industry climate change survey allows us to track changes in insurers' stated investing policies, providing an additional measure of the impact of treatment (NAIC, 2023). Fourth, the setting avoids contemporaneous event issues of policy changes. The rule wasn't bundled with other changes; no new climate-related disclosure regulations were passed in the U.S. during our analyses, nor was there any climate-related portfolio information released by a third party such as Morningstar. Lastly, the rule was not subject to a prolonged legislative process where lobbying efforts could have altered the outcome because the CDI has the sole authority to regulate insurance companies licensed in California and implemented the rule with a relatively short timeline and limited comments. While our setting has several empirical advantages, we caveat the results because our setting potentially limits the generalizability of the results, as we examine one type of institutional investor and one disclosure mandate with a specific disclosure channel.

To test the effect of the mandatory disclosure, we adopt a difference-in-differences (DiD) approach surrounding the 2016 enactment of the California Rule. We focus our analysis on bonds because bonds are a large portion of insurance portfolios, it avoids combining equity and other securities with vastly different characteristics, and it allows us to use par values to avoid perceived changes in portfolio securities due to market fluctuations. We compare the difference between pre- and post-period fossil fuel investments of disclosing firms with the fossil fuel investments of non-disclosing firms and find that the disclosure results in statistically and economically significant divestment in the post-period. Specifically, the average California-licensed insurer held \$44M in bond fossil fuel investments in the pre-period and divested in the post-period approximately 20 % of its fossil fuel holdings, or \$9M on average. The results are robust to different matching procedures, fixed effects structures, insurer characteristics, and fossil fuel commodity prices.

In addition to the significant on-average result, we observe meaningful variation in the magnitude of divestment, e.g., 27 % of firms did not divest or increased exposure over the period. We examine the factors associated with the decision to divest. Using only firms impacted by the disclosures, we first examine if the pre-policy portfolio mix affects divestment decisions and find that firms with

[1] Throughout the paper we refer to "divesture" as either a full or partial reduction in investments in fossil fuel companies. While colloquial usage of the term often seems to refer to a complete divesture resulting in zero holdings, the definition of the term also includes a decline in holdings.

[2] In the U.S., the SEC's proposed "Enhanced Disclosures by Certain Investment Advisers and Investment Companies about Environmental, Social, and Governance Investment Practices" requires environment-focused funds to disclose the greenhouse gas emissions of the firms they invest in (SEC, 2022).

[3] As this quote shows, many of these initiatives are designed to provide information beyond financial information to a broad set of stakeholders, not just to capital providers. In fact, the European Parliament has stated "Financial and sustainability reporting will be on an equal footing and investors will finally have access to reliable, transparent and comparable data" (European Parliament, 2024).

[4] For example, our paper differs from the concurrent working papers in the following ways. While there was no concurrent disclosure regulation that impacted investors in our setting, in Messonier and Nguyen (2021) the regulation they study (Article 173 in France) was passed at the same time as the Paris Climate Accord. In Dai et al. (2024), the disclosure regulation only applies to those firms that self-identify as sustainability funds and in Gehricke et al. (2025) only the top 200 largest corporate issuers, insurers, creditors, and fund managers face the disclosure requirements.

[5] Access to the data is based on a proprietary database. While it may be available to sophisticated investors, the costs to obtain and process the data would likely be prohibitive for consumers or individual investors. Further, the rule required firms to classify securities as being fossil fuel investments – prior to these disclosures investors would have had to create their own classifications.

above-median fossil fuel bonds are the ones mostly divesting, consistent with efforts to conform to the industry benchmark.[6]

Next, we turn to investigating regulatory pressure. In California, the commissioner approves both rate increases and the right to do business in the state. When the rule was announced, observers argued that insurers would "fear retribution" if they did not appease the regulator by divesting fossil fuel securities, particularly given the commissioner's stated preference for divestiture (Greenhut, 2016).[7] To examine whether the reduction is a response to, direct or feared, increased regulatory pressure, we investigate whether firms with a higher percentage of their premiums in California or those that are headquartered in California are more likely to divest than other firms. Either attribute would provide the insurance commissioner with a greater ability to impact the firm. However, we find no relation with either measure, suggesting direct regulatory pressure is not a primary driver of the decision to divest.

Failing to find a direct regulatory link, we investigate outside stakeholders' influence on the decision to divest. We begin by examining whether environmental activists have previously targeted the firm. We find that previously targeted firms are more likely to reduce their holdings once disclosure is required. Next, we examine whether public firms, which prior research has shown can face environmental stewardship pressure from investors, are more likely to divest.[8] Consistent with investor pressure, we find that public firms are more likely to divest, and that within public firms, those that are larger (thus more visible) divest more. Finally, we examine size as a determinant as well and find that firm size is positively correlated with divestment. These tests suggest that disclosure is most likely to impact firms with outside stakeholders who could use the information to advocate for change.[9] These findings on the importance of external stakeholders are consistent with the goal of the commissioner to "… make this new information public so that investors, policyholders, regulators, and the general public can know the extent to which insurance companies are invested in the carbon economy" (Jones, 2016a). The results suggest it is the disclosure itself and not underlying regulatory pressure that causes insurers to divest.

A unique feature of our setting is that the disclosure mandate ends in 2019, allowing us to observe whether the portfolio changes were temporary. Looking at the post-period, we find that the insurers who made changes to their investment portfolios generally do not revert to their pre-policy holdings of fossil fuel investments. Using disclosures of investment policies required in several states, we show that divesting firms added climate-related policies. Further in the post-period, insurers with specific climate investment policies do not revert to their pre-disclosure investment policies. This suggests that the permanent changes in insurers' fossil fuel holdings are potentially due to the concurrent changes in the insurers' formal investment policies.

We next investigate whether firms engaged in window dressing by selling fossil fuel securities to non-impacted subsidiary insurers owned by the same parent company, or divesting prior to year-end and buying back the securities at the beginning of the following year. We find no evidence of either behavior. Instead, ~90 % of divested securities are bought by institutions outside of the insurance industry. However, we find robust evidence that fossil fuel divestment comes at a cost. Disclosers that divest experience lower investment returns compared to non-disclosers.

This paper makes several contributions. First, the paper's findings provide a unique insight into the effect of mandatory disclosure on institutional investors. Prior papers have studied the effect of investors on operating firm disclosure (Bird and Karolyi, 2016; Tsang et al., 2019; Abramova et al., 2020), but there is little evidence on how disclosure regulation targeted at investors affects their holdings or investment policies. This paper also adds to the literature on mandatory nonfinancial disclosure and, more specifically, to the growing literature on environmental disclosure, of which climate risks are one of the most significant components. While prior research has focused on the effects of mandatory disclosure on firms' carbon emissions and pollution (Downar et al., 2021; Jouvenot and Krueger, 2019; Tomar, 2023), it is unclear whether investors will respond similarly to disclosure regulation.[10] This is one of the first papers to examine the effects of mandatory environmental disclosure on institutional investors (Messonier and Nguyen, 2021; Dai et al., 2024; Gehricke et al., 2025). By showing the impact of external stakeholders, the paper highlights the role of public disclosure in influencing investment decisions.

Second, this paper provides new insights into the burgeoning literature on climate finance, which is the study of local and global financing of public and private investment that seeks to support the mitigation of and adaption to climate change (UNFCC: The United Nations Framework Convention on Climate Change, 2022; see Hong et al. (2020) for a review). Previous studies have focused primarily on voluntary actions by institutional investors (Choi et al., 2021; Humphrey and Li, 2021; Ceccarelli et al., 2021, 2022; Ilhan et al.,

[6] While the decline is also consistent with regression to the mean, we do not observe a similar regression upward for firms below the median. Further, our subsequent cross-sectional results do not support that interpretation.

[7] For example, the Insurance managers may have feared the commissioner would block rate increases or make the process more burdensome unless they show reductions in their fossil fuel investments.

[8] For example, both Dyck et al. (2019) and Chen et al. (2020) find that institutional investors may pressure for environmental improvements. Similarly, Brownen-Trinh and Orujov (2023) show retail investors may influence socio-political decisions by firms.

[9] This evidence is consistent with insurers statements in the NAIC Climate Risk Survey. For example, one insurer explained, "our initial risk assessment showed there are only minor risks to [our] business from climate change; our main risks are reputational."

[10] There are several reasons why investors may respond differently to environmental disclosures compared to issuers (non-financial firms). First, issuers may receive benefits for their decarbonization. They may be able to market their more sustainable products (Dai et al., 2021; Christensen et al., 2023) or they may save on operating costs or have more stable energy costs (Al Khaffaf et al., 2024). The benefits for investors are less tangible, especially for insurers where the composition of the investment portfolio is likely not a significant factor in consumers' purchase decisions. Another key difference is that investors and fund managers need to balance risk and reward when they construct their portfolio and fossil fuel securities can be a valuable source of diversification with returns uncorrelated with the rest of the portfolio and no clear substitutes. Another key difference between investors and issuers is how quickly (or slowly) each can respond to mandatory disclosure. For issuers, decarbonization would likely require substantial operational changes to their business. For investors, divesting can be much quicker.

2021). This paper diverges from those by focusing on the effect of mandatory disclosure. Additionally, our analyses on fossil fuel securities are an answer to Hong et al. (2020) call for more research on divestments and stranded assets.

Third, this paper provides empirical evidence relevant to an ongoing policy debate. Regulators in several countries have proposed/implemented climate-risk disclosure for institutional investors (Duran, 2021; Halper et al., 2022; Teu, 2021; Treacy et al., 2022). Some argue direct regulations such as taxing emissions (Lester and Olbert, 2024) or mandatory divestments are preferable (Carlin, 2021; Reitmeyer, 2022), while others are concerned about the significant costs of complying with the disclosures (ERM, 2022).[11] This paper provides initial evidence that mandatory disclosure can be effective in changing investors' fossil fuel holdings and investment policies by engaging outside stakeholders as monitors. To the extent that the disclosures led to pressure from those outside stakeholders, it provides evidence consistent with the argument that the "voice" of engagement can be effective in changing behavior, while our findings that the fossil fuel company's borrowing costs do not change with divestment provides evidence that "exit" (divestment and boycott), at least in this setting, may be ineffective in changing firm behavior (Broccardo et al., 2022).[12]

Lastly, this paper investigates mandatory disclosures created by a single state and outside the normal reporting regime. There is a growing movement of state-mandated environmental disclosures, including a broader set of rules from California as well as potential rules in New York and Illinois (Ring, 2024). This is one of the first papers to examine the impact of rules from these non-standard setting bodies. Our evidence of change in firm behavior suggests that future state-based rules may also have an impact on economic operations.

2. Background, conceptual framework, and sample

2.1. Institutional background

California's insurance market is the largest in the U.S. and the 4th largest in the world (Insurance Journal, 2018; CDI, 2022). The state's insurance market distinguishes itself through its size and by the significant authority vested in its commissioner. In the late 1980s, the role was changed from government-appointed to elected by voters at large and given sole authority over insurance rates. This freed the commissioners from oversight by the governor and state legislator, but also created political incentives for the regulators holding these positions. Subsequent commissioners have often used their power to achieve specific policy aims.[13]

In January 2016, then-commissioner Dave Jones ordered insurance companies licensed in the state to disclose their fossil fuel securities. The rule defined "fossil fuel" security as any security issued by a company that extracts oil, gas, and coal or an electric utility that uses fossil fuels to generate electricity (Jones, 2016a).[14] In addition to the mandatory disclosure, the commissioner argued that holdings in fossil fuel companies represent a risk to policyholders as he felt those assets are likely to decline in value. Accordingly, he asked insurance companies to voluntarily divest from investments in thermal coal.[15] He specifically asked that they stop making new investments, refrain from renewing existing investments, and sell or withdraw from existing investments in any company that generates 30 % or more of its revenue from the mining or use of thermal coal. At the time of the rule's announcement, California-licensed insurers collectively held close to $500M in debt securities that meet these criteria.

Jones' stated goal was to make all stakeholders aware of the insurers' fossil fuel investments. Pundits were quick to argue that the rule was politically motivated (Greenhut, 2016; Lehman, 2016), as the initial announcement came just a few days before the commissioner's speech at a UN climate event (Dentons, 2016).[16] The rule was met with vocal opposition from insurers, fossil fuel companies, and other insurance commissioners. Commissioners from six oil-producing states sent a public letter to the California

[11] A recent survey of 35 institutional investors with over $7 trillion in assets under management found investors on average spend $1,333,000 annually to collect, analyze and report climate data (ERM, 2022). With 745 active institutional investors in the United States and Europe, the costs of compliance are near $1 billion annually (Schroders, 2023).

[12] While Broccardo et al. (2022) model a single investor and an operating company, our setting has an additional layer of ownership. First, the insurance company has outside investors. To the extent that investment in emission producing companies as part of their business practices is considered an environmental negative, Broccardo et al. (2022) would suggest that firms with a higher probability of investor "voice" would reduce such holdings. Consistent with this, our findings indicate a broader and previously more vocal group of such stakeholders leads to a decline in a negative environmental activity (stopping investment in an activity related to emissions). Second, the insurance companies are investors themselves, and we see that when they divest there is no change to the fossil fuel company's borrowing costs. This suggests divestment had a limited effect on the fossil fuel companies, again consistent with the predictions by Broccardo et al. (2022). With that said, it is not surprising that the insurers opted to divest instead of pressuring the fossil fuel companies to reduce their greenhouse gas emissions. The CA commissioner made it clear that he wanted divestment and the disclosure narrowly focused on the amount invested in fossil fuel companies (Jones, 2016a).

[13] For example, in 2010, then-commissioner Steve Poizner floated a plan to force all insurers who operate in California to divest investments in any multinational companies that do business in Iran due to his view that they are a sponsor of terrorism (Greenwald, 2010).

[14] Responding to questions from insurers, Jones clarified the definition as securities issued by companies who i) generate 50 % or more of their revenues from oil and gas, ii) generate 30 % or more of their revenue from thermal coal or iii) utilities that generate 30 % or more of their electricity from thermal coal or that generate 50 % or more of their electricity from the combustion of oil or natural gas (Jones, 2016b).

[15] Thermal or steam coal is a term used for non-meteorological coal. This is the coal used to generate electricity.

[16] Elected executive officials in California, including the insurance commissioner, are limited to two terms.

commissioner asking for him to cease the ruling (Doak et al., 2016).[17] However, the California commissioner choose to keep disclosure regulation in effect and it remained until the end of Jones' term in 2019.

The California Department of Insurance (CDI) collected the fossil fuel holdings information from the insurers, summarized it, and then published the holdings at the insurer level on the CDI website.[18] Appendix B provides a screenshot of the website which displays information such as the insurer's NAIC number, company name, California premiums, assets under management, fossil fuel investments including coal, percentage of fossil fuel investments to assets under management, value of coal investment over the CDI threshold, and value of fossil fuel investments over the CDI threshold. Prior to the mandated disclosure, market participants could access the NAIC filings to view the details of an insurer's raw investment holdings. However, there were no classifications provided for fossil fuel investments, so investors would have had to research each holding after creating their own criteria. Following the disclosure mandate, the CDI website provided all companies' calculated holdings and allowed users to easily sort insurers by their total fossil fuel holdings or their holdings as a percentage of the portfolio.[19] Several media outlets accessed the public disclosure and reported on it, expanding the reach of the information to those who did not access the CDI site on their own (Reuters, 2016; Bloomberg, 2016).

2.2. Motivation

While the new regulation did not require mandatory divestment or other changes in investing choices, its requirement for disclosure likely caused insurers to both anticipate potential stakeholder responses and to monitor actual responses as they occurred. The actual or anticipated responses by the users could have fed back into the insurers' decisions, resulting in changes in investment or other real actions (sometimes referred to as an "action cycle") (Tomar, 2023; Hombach and Sellhorn, 2019; Weil et al., 2006).

Several factors make it possible for an "action-cycle" to lead to divestment in this situation. First, pressure from external stakeholders could harm the reputation of the insurer and its managers (Dewatripont et al., 1999). The insurance industry relies heavily on trust and reputation, and over the last decade, fossil fuel investments have faced growing criticism due to their contribution to climate change and associated environmental and social risks (Vetter, 2020). As Hong et al. (2020) summarize, "[E]nergy companies have become the new 'sin' stocks facing divestment campaigns and lawsuits from shareholders … The divestment and legal campaigns are similar to what tobacco companies faced a generation ago …" Insurers may anticipate the negative publicity and divest as a result. In support of this incentive in our setting, a recent survey of institutional investors finds that the strongest motive for investors not to invest in fossil fuel companies is the protection of the investor's reputation (Krueger et al., 2019).

Regulatory pressure may also motivate insurers to divest (Cahan, 1992). As mentioned earlier, rates for California-licensed insurers are approved by the commissioner. Insurers may fear the commissioner will hinder rate increases unless they reduce their fossil fuel holdings. As Greenhut (2016) explained at the time of the rule's announcement,

> "Insurers are in something of a bind when it comes to rebutting any insurance commissioner. The commissioner has so much authority over insurers that they rightly fear retribution when it comes to, for example, rate hike requests. So while the commissioner and his supporters say the divestment request is voluntary, it's easy to see why insurance companies might believe there is some duress involved."

On the other hand, there are several reasons why the California Rule may not result in any change to the investment company's holdings or investment policies. First, divestment may hurt the fund's performance by limiting diversification and forcing the investors to sell securities at less than their carrying values (Shleifer and Vishny, 2011). If this compromises the insurers' financial health or results in poor investment returns, the company could face lawsuits from shareholders, policyholders, or other stakeholders. Similarly, the managers are compensated based on the portfolio's returns, and any poor performance due to divestment would be directly felt by them.

Also, it seems unlikely that the requirement to disclose would change the insurance companies' assessments of the underlying impact of climate risk on their investment policies. Insurers are already at the forefront of insuring against weather-related events (Bank of England, 2015) and are likely taking climate change and its implications into account with their investment strategies, which are known for being conservative (Grundl et al., 2016). Further, divestment is rarely used by investors. A survey of institutional investors by Krueger et al. (2019) identified 12 possible approaches to deal with climate risk; the least frequently used tool by investors is to divest problematic portfolio firms. This is shown by Ansar et al. (2013) as well, who found that during a three-decade divestment campaign against tobacco companies, only about 80 organizations and funds ever sold stock to support the campaign.

On balance, there are reasons to expect divestment as well as reasons to anticipate the rule will have no impact on many (or all) firms. Thus, our first set of analyses will examine divestment in the sample of firms. However, it is also likely that firms may face differing incentives to divest, leading us to consider cross-sectional factors that may impact the decision to modify the investment portfolio. Firms that face a greater potential economic impact if they ignore the commissioners' request for divestiture may be more likely to divest. Similarly, those who face greater pressure from external stakeholders may find the disclosures are used to reinforce current divestiture campaigns or to identify firms as targets of new campaigns.

[17] Along with arguments the commissioner's ruling was politically motivated, others objected to the disclosure itself because it would cause unwarranted financial loss for insurers. Conversely others argued the disclosure did not go far enough and would be ineffective as a result (Greenhut, 2016).

[18] Holdings data can be viewed here: https://interactive.web.insurance.ca.gov/apex_extprd/f?p=260:1.

[19] The main CDI website receives approximately 1.8M views each year (Hyperstat, 2023).

In considering these two potential pressures to divest, it is important to note that in the case of direct regulatory pressure, the regulator could apply the pressure without requiring a broader public disclosure. In that case, the disclosure would simply be a way for the commissioner to highlight the impact of their regulatory pressure. That is, divestiture could have occurred without the disclosure.[20] However, in the second case, creating external stakeholder pressure, the disclosure would act as a tool for empowering or attracting external stakeholders, making the disclosure a more direct and essential part of the divestiture process.

2.3. Sample and data

2.3.1. Sample selection

We construct the sample for our main analyses using security-level holdings from the National Association of Insurance Commissioners (NAIC), the insurance industry's national regulator and standard setter. Insurers are required to file their investment portfolio with the NAIC annually.[21] We gather the end-of-year portfolio holdings (i.e., EOY portfolio) from Schedule D – Part 1 and information about annual purchases and sales from Schedule D – Parts 3 and 4 (i.e., transaction data) for all property and casualty insurers from 2014 to 2021.[22] The EOY portfolio reports all credit obligations owned by the insurers as of December 31st for each year, including corporate bonds, municipal securities, bank loans, treasury notes and treasury bonds, along with mortgages and various asset-backed securities. The transaction data covers all the insurers' buys and sales of stocks and bonds for each year. This paper focuses on bond holdings specifically because i) bonds are the majority of insurers' investment portfolios,[23] ii) they allow us to use par values, thus avoiding the impact of market values in causing perceived changes in portfolios, and iii) they allow us to avoid combining securities with vastly different characteristics.

The EOY portfolio includes information including the CUSIP, the name of the issuing entity when the security was acquired, acquisition costs, the effective interest rate, the unrealized gains or losses, and the interest paid and due for that year. It also includes the par and fair value of the security at the end of the fiscal year. The transaction data includes identifying information about the security along with the date the security was acquired or sold and the third party that facilitated the transaction. We combine the EOY portfolio with the transaction data to create quarterly-level portfolio holdings. We identify financial information about each insurer from the Bureau van Dijk Orbis dataset, which includes the gross, net, and earned premiums, total assets, and net investment income for each insurer. We add summary information from the National Association of Insurance Commissioners (NAIC), including the name of the insurer's parent company and the state in which the insurer is domiciled.

We classify an insurer as licensed in California if it is listed on the CDI's website.[24] We restrict the sample to only insurers with a complete set of information. This selection process results in our primary sample consisting of 43,820 quarterly observations containing 1932 unique insurers from a wide range of states. See Table 1, which details the sample selection process.

2.3.2. Measuring Fossil Fuel Investments

Our primary variable is the fossil fuel investments held by each insurer. We measure *Fossil Fuel Securities/Tot Bonds* as the total fossil fuel investments held by the insurer scaled by the value of the insurer's total bond portfolio. Par value is set when the bond is issued and does not change over time.[25] Any changes in this amount are then due to the insurers buying or selling the securities. Alternatively, we could have used market values. However, those amounts include both actual changes in holding and changes in value due to market changes, introducing noise into a measure of divestment. In untabulated results, we use the security's fair value instead of par value and find statistically and economically similar results.[26]

Rather than providing a specific list of fossil fuel securities or industries, the CDI defined fossil fuel securities as publicly traded securities where the company generates a significant portion of its revenues from oil and gas, coal, or utilities that generate a significant percentage of its electricity from fossil fuels (Jones, 2016b). Firms were required to analyze their holdings to determine whether any securities met these criteria. We use the classifications made by the firms in their reports to identify fossil fuel securities. We use the list of fossil fuel securities for each insurer per the CDI website, identify the issuing companies, and then label a company as

[20] Some might argue that even in that case the public disclosure would serve as a commitment mechanism for the commissioner. That is, by requiring disclosure the commissioner commits to letting others assess their effectiveness in causing firms to divest as well as signals to the firms that the commissioner cannot withdraw pressure given the public disclosure of the outcomes.

[21] We received the data directly from the NAIC. For stakeholders to access insurers' holdings before the California Rule they could either purchase the Schedule D filings from the NAIC or view the data from a service provider like Bloomberg. Both options are costly and likely deterred stakeholders from obtaining the holdings prior to the California Rule.

[22] The sample starts in 2014 because that's the first year the NAIC has compiled the Schedule D – Part 1 data.

[23] Data from the NAIC on the $2.3T investments held by U.S. property and casualty insurers at year-end 2020 shows that over half are held in bonds (NAIC, 2021).

[24] An insurer is licensed to do business in the state when the insurer has met the legal requirements and obtained the necessary authorization from the state's insurance regulatory authority to conduct business within the state. There are financial and solvency requirements, compliance issues and product approvals just to name a few of the requirements.

[25] As discussed earlier, the ability to use par value is one advantage of focusing on bonds over other investments, such as equity. In addition, bonds make up the majority of most investment portfolios and focusing on just one type of security reduces measurement issues of combining securities.

[26] While market changes in general could introduce noise into the measure, if divesture by some insurers negatively impacts market values, then use of market values could result in a conclusion of divesture by other firms due solely to this market value decline. Given that such impact would only influence the fossil fuel investments, it would bias our results towards finding divesture.

Table 1
Sample selection.

Main Sample	
Step	Observations
All monthly observations covered by the NAICS data	295,308
Exclude observations that do not have Bureau van Dijk Orbis information	(86,437)
Exclude firms that do not have observations in both the pre- and post-periods.	(1664)
Exclude firms that do not have certain NAIC control variables	(2652)
Exclude firms that do not have net investment income	(7358)
Exclude observations after 2019	(46,930)
Exclude observations NOT in March, June, September or December	(106,447)
Total quarterly observations	43,820

Notes. This table outlines the sample selection process.

a fossil fuel company if five or more insurers identify the company as a fossil fuel company.[27] This ensures we are identifying generally agreed upon fossil fuel companies. We then label all the securities issued by those companies as fossil fuel securities and aggregate the fossil fuel securities owned by each insurer per quarter to arrive at the total fossil fuel investments. We scale by the par value of the insurer's total bond portfolio to create the main outcome variable.

2.3.3. Descriptive statistics

Table 2 presents the descriptive statistics of the key variables used in our analysis. Panel A presents the variables for the main sample. The average insurer has a total bond portfolio of just under $500M. On average, fossil fuel investments make up approximately 6.8 % of the insurer's bond portfolio. Utilities account for 3.7 % of the portfolio, coal accounts for 0.1 %, oil and gas accounts for 2.5 %, and other fossil fuel securities account for the remaining 0.5 %. There are 314 California-licensed insurers (*Treat*), with the remaining 1618 insurers licensed outside the state.

3. Main results

3.1. Effect of the California rule on manager's porfolio choices

3.1.1. Main specification

We begin our empirical analyses by estimating the effect of mandatory disclosure on the investment company's holdings of fossil fuel investments. Prior to our full analysis, Fig. 1 shows the mean (Panel A) and median (Panel B) percent of California insurers' fossil fuel investments over time by plotting holdings at the end of each quarter. Both the mean and the median show a decline in fossil fuel holdings over the duration of the rules. While declines continue somewhat after the rule ended in 2019, the largest declines in both mean and median are during the treatment time.[28]

As a final univariate investigation, we examine changes in fossil fuel investment by conditioning on relative investment levels prior to the disclosure rule. In Fig. 2, we examine investment by firms below the median (Panel A) and above the median (Panel B) in pre-rule fossil fuel holdings. We see a small decline in fossil fuel investments for those already below the median (from about 5 % to slightly above 4 %) but a larger decline in those initially above the median (from 14 % to approximately 11 %). The decline observed in the higher investment group would be consistent with a regression to the mean. However, the modest decline in the below-median group is inconsistent with a regression to the mean explanation. Overall, the univariate results indicate that the new rule was associated with some degree of investment and had a greater influence on those with a larger ex-ante fossil fuel exposure. In later tests, we return to examining the cross-sectional decisions to better understand other determinants.

Moving beyond this univariate analysis, we use a difference-in-differences research design with insurers licensed in California considered "treated" while all insurers in other states are in the untreated control group. The sample period starts in 2014 and ends in 2019, when the commissioner's term ends. Because the policy became effective for insurers on December 31, 2016, we use that quarter-end date as the start of the post-period and all prior quarters as the pre-period. Eq. (1) represents our model for testing the effects of mandatory disclosure (firm and time subscripts are omitted):

$$\textit{Fossil Fuel Securities} / \textit{Tot Bonds} = \alpha + \beta \textit{Treat} \times \textit{Post} + \textit{Treat} + \textit{Post} + \textit{Controls} + \lambda + \theta + \varepsilon \tag{1}$$

We estimate Eq. (1) using OLS regression and heteroscedasticity-robust standard errors clustered at the state-year level, as

[27] We manually inspected those securities where just a few insurers said the issuer was a fossil fuel company and found several financial firms including J.P. Morgan, General Electric, GE Capital and Lowes Corp. among others. While these firms may underwrite and finance fossil fuel companies, they are not pure fossil fuel companies. By requiring five insurers or more to agree that the security comes from a fossil fuel company we reduce the impact of idiosyncratic classifications as well as those that are just errors of a small number of insurers.

[28] While the figure appears to show a small decline beginning in 2016 prior to the rule being enacted, in untabulated results we find the levels of fossil fuel securities between the first two quarters of 2014 and the last two quarters of 2016, just prior to the onset of the disclosure mandate, are statistically equivalent.

Table 2
Descriptive statistics.

	Ind/Cont	N	mean	sd	p25	p50	p75
Dependent Variables:							
Fossil Fuel Investments	C	43,820	0.068	0.084	0.019	0.055	0.092
Utilities	C	43,820	0.037	0.057	0.000	0.026	0.052
Coal	C	43,820	0.001	0.004	0.000	0.000	0.000
Oil & Gas	C	43,820	0.025	0.052	0.000	0.016	0.035
Other	C	43,820	0.005	0.013	0.000	0.000	0.005
Test Variables:							
Treat	I	43,820	0.171	0.377	0.000	0.000	0.000
Post	I	43,820	0.617	0.486	0.000	1.000	1.000
Control Variables:							
Gross Premiums	C	43,820	17.27	2.26	15.93	17.65	18.96
Total Assets	C	43,820	17.80	2.03	16.61	18.01	19.25
Investment Income	C	43,820	0.312	0.459	0.040	0.079	0.858
Additional Variables:							
Fossil Fuel Investments ($M)	C	43,820	44	207	0	3	14
Total Bond Fair Value ($M)	C	43,820	499	2220	15	59	213

Notes: This table presents the summary statistics for the main sample with quarterly observations. Variable definitions are in Appendix A.

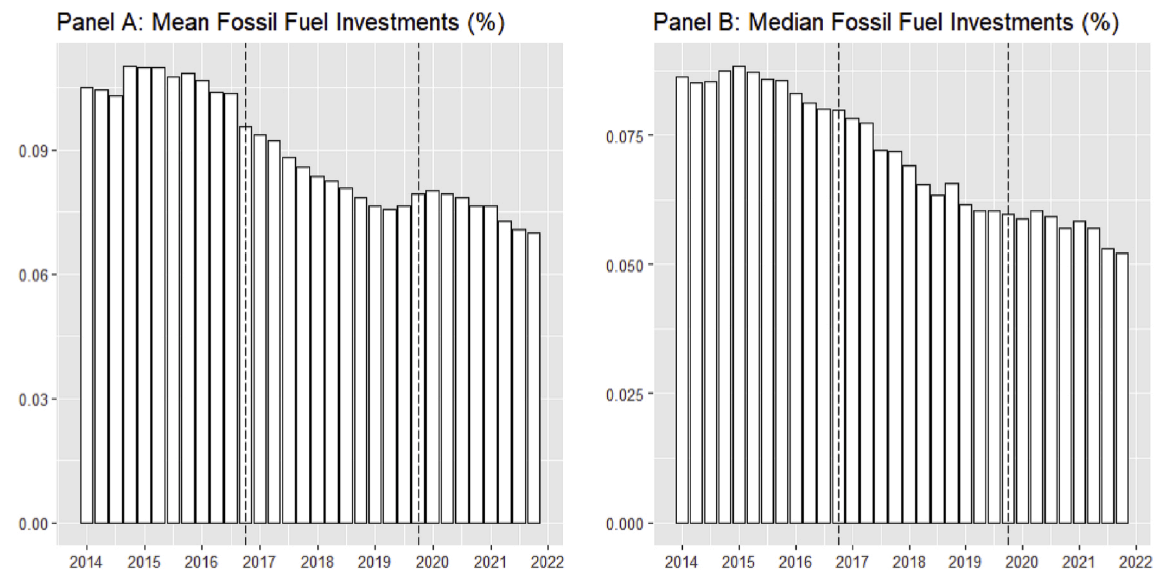

Fig. 1. Mean and Median Fossil Fuel Holdings Over Time
Notes. These figures show the mean (Panel A) and median (Panel B) fossil fuel holdings over time for CA-licensed insurers. Holdings are measured as *Fossil Fuel Securities/Tot Bonds* which is the par value of fossil fuel investments divided by the total par value of the firm's bond portfolio. The dotted lines denote the start and end of the disclosure mandate.

treatment is based on the state where the insurer is licensed. The dependent variable *Fossil Fuel Securities/Tot Bonds* is the par value of the insurer's fossil fuel investments divided by the par value of the total bond portfolio (see the section "Measuring Fossil Fuel Investments" for more details). The indicator variable *Treat* distinguishes between insurers licensed in California and all other U.S. insurers.

In our main analysis, we restrict the California-licensed insurers to only those with fossil fuel securities immediately prior to the onset of the regulation, allowing us to focus on firms that have an opportunity to divest. Our results are robust if we remove this restriction (see Table 11). *Post* is a binary variable equal to one for the period after the announcement of the California Rule (from 12/31/2016 onwards) and zero for the prior period. *Controls* are a vector of control variables to account for observable differences that may be correlated with the investor's decision to invest in fossil fuel securities. For example, size may be correlated with the decision to

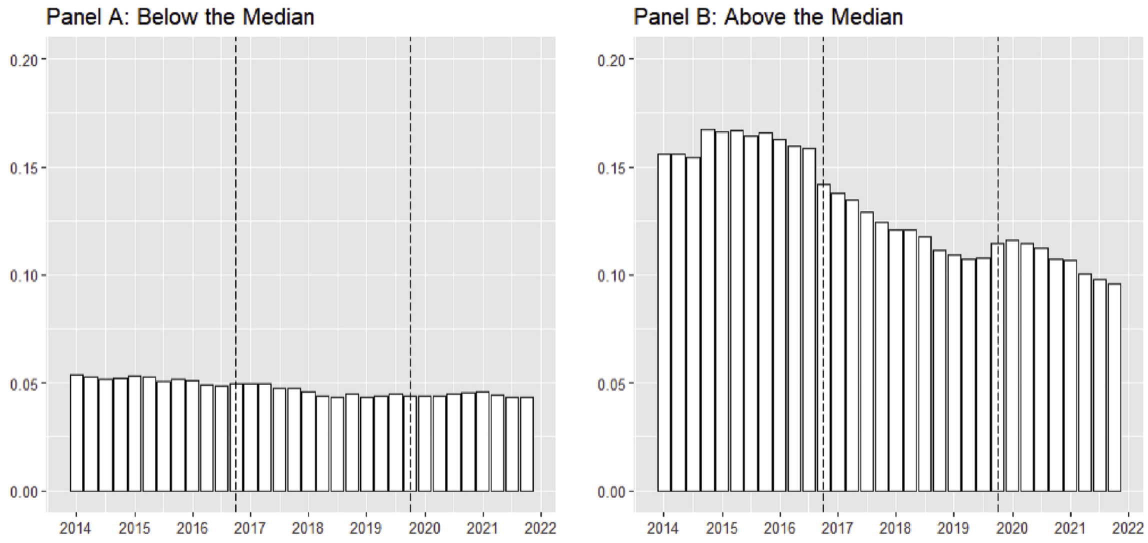

Fig. 2. Fossil Fuel Investment Over Time by Pre-Rule Fossil Fuel Holdings
Notes. These figures show the change in fossil fuel investments over time for California-licensed insurers for those below the median (Panel A) and above the median (Panel B) in pre-rule holdings of fossil fuel securities. We measure *Fossil Fuel Securities/Tot Bonds* as the par value of fossil fuel investments divided by the total par value of the firm's bond portfolio. The dotted lines denote the start and end of the disclosure mandate.

divest as larger firms have a wider range of investment options (Grundl et al., 2016) and are more likely to face pressure from their broader range of stakeholders (Darnall et al., 2010). We follow prior papers and use both *Gross Premiums* and *Assets* as proxies for size (Anantharaman, 2017). We further use the percentage of income earned by investment income (*Investment Income (%))* as greater dependence on investment income likely affects the investment managers' investment risk. Certain insurers may be less willing to divest if they fear divestment will hurt their returns. To control for both time-invariant unobservable differences in firm characteristics and time-varying unobservable factors, we employ firm- (λ) and quarter-year (θ) fixed effects, respectively. Appendix A provides variable descriptions for all variables used.

Table 3 provides estimates for the effect of mandatory disclosure on insurers' fossil fuel investments. We find that treated firms (i.e., firms licensed in California and thus required to disclose investments) reduce their holdings in fossil fuel securities, consistent with mandatory disclosure, leading to an on-average voluntary divestment of fossil fuel investments. Column (1) includes only firm and quarter-year fixed effects as control variables. This column mitigates the concern that including covariates affected by the treatment may impact causality claims (Gormley and Matsa, 2014). The main coefficient of interest is the interaction term, which is negative and significant (coef. = −0.013, t-stat = −5.64).[29] Column (2) adds controls for firm characteristics and finds similar results. In both columns, *Fossil Fuel Securities/Tot Bonds* decrease by approximately 1.3 %, which corresponds to a decrease of approximately 20 % (0.013/0.068) of the average scaled fossil fuel investments.[30] With the average fossil fuel investments of $44M, the effect is equivalent to a divestment of approximately $9M on average. The specification in Column (2) is the main specification used throughout the remainder of the study. The results are similar when we further control for *State* × *Year* in column (3), which suggests the findings are not driven by unobservable time-varying factors pertaining to the state. In column (4), we control for changes in the annual prices of fossil fuel commodities, including coal, natural gas, and oil to ensure the results are not due to macroeconomic changes in the commodities. We find similar results to the prior specifications.[31]

[29] The result is robust in a specification with no firm or quarter-year fixed effects (see Table 11).
[30] The on average 20 % reduction in fossil fuel holdings is similar to the reductions found in other papers for issuers and investors. Downar et al. (2021) find UK (non-investment) firms reduce their greenhouse gas emissions between 14 and 18 % relative to a control group with the onset of a disclosure mandate. Messonier and Nguyen (2021) find French institutional investors reduce their holdings of fossil fuel securities by 32 % on average. Dai et al. (2024) find the EU-market funds claiming to invest based on sustainability criteria have a 10 % reduction in the greenhouse gas emissions of their portfolio following the introduction of the EU SFDR.
[31] The constant term is the average percentage of insurers' bond portfolios held in fossil fuel securities when all other regressors are equal to zero (Breur and deHaan, 2024).

Table 3
Effect of mandatory disclosure on fossil fuel investments.

Dep. Var =	*Fossil Fuel Securities/Tot Bonds*						
	No Matching				Entropy Balanced Matching		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Treat × Post	−0.0130*** (-5.64)	−0.0130*** (-5.64)	−0.0128*** (-5.77)	−0.0130*** (-5.64)	−0.0131*** (-5.56)	−0.0120*** (-6.08)	−0.0131*** (-5.57)
Gross Premiums		−0.000387 (-0.60)	−0.000612 (-0.91)	−0.000386 (-0.60)	0.00204** (2.05)	0.000802 (0.76)	0.00203** (2.05)
Assets		0.000803 (0.82)	0.00162* (1.69)	0.000813 (0.83)	−0.00164 (-1.30)	0.000832 (0.76)	−0.00161 (-1.28)
Investment Income (%)		0.00262** (2.52)	0.00231** (2.07)	0.00263** (2.52)	0.000335 (0.15)	0.000955 (0.45)	0.000339 (0.15)
Post			−0.00194 (-1.20)	−0.00175 (-1.00)		0.0000306 (0.01)	0.000267 (0.11)
Coal Prices				−0.0251* (-1.74)			−0.0160 (-0.71)
Natural Gas Prices				0.0103*** (4.00)			0.00841** (2.24)
Oil Prices				−0.00843*** (-4.71)			−0.00997*** (-3.79)
Constant	0.0690*** (123.67)	0.0606*** (3.82)	0.0510*** (3.36)	0.0613*** (3.87)	0.0746*** (3.53)	0.0521*** (3.10)	0.0741*** (3.50)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	No	No	No	Yes	No	No	Yes
Quarter-Year FE	Yes	Yes	No	No	Yes	No	No
State-Year FE	No	No	Yes	No	No	Yes	No
N	43,820	43,820	43,820	43,820	43,820	43,820	43,820
R-squared	0.818	0.818	0.824	0.818	0.840	0.851	0.839

Notes. This table presents an estimate of the effect of the California Rule on insurers' fossil fuel investments. The sample consists of quarterly observations. The dependent variable *Fossil Fuel Securities/Tot Bonds* is the par value of fossil fuel investments divided by the total par value of the firm's bond portfolio. *Treat* × *Post* is a binary variable that equals one for California-licensed insurers after the announcement of the California Rule (i.e. 12/31/2016 onwards). Columns (1), (2), (4), (5) and (7) report results controlling for firm and quarter-year fixed effects. Columns (3) and (6) report results controlling for firm and state-year fixed effects. Columns (5) through (7) use entropy-balanced matching on gross premiums, assets, bond size and the average value of the fossil fuel investments held by the insurer in the pre-period. Variable definitions are in Appendix A. Standard errors are clustered at the state-year level. T-stats are displayed in parentheses. *, **, and *** indicate significance levels of 10 %, 5 % and 1 %, respectively.

To address potential selection concerns, we also provide results based on entropy-balanced matching, which ensures there are equal covariate balances based on observable characteristics (Hainmueller, 2012). We match on four characteristics: gross premiums, assets, the value of the bond portfolio, and the average value of the fossil fuel investments held by the insurer in the pre-period. Column (5) presents the results for the main specification. Column (6) controls for *State* × *Year* interactions, and column (7) additionally controls for fluctuations in commodity prices. Across all three columns, the coefficient on the interaction term is negative and significant, suggesting that mandatory disclosure is associated with fossil fuel divestment.[32]

[32] Insurers may be investing in green bonds issued by fossil fuel companies either before or during the disclosure period. Green bonds are issued by companies with a high carbon footprint to fund projects with positive environmental impacts, such as renewable energy, energy efficiency, clean transportation, sustainable water management, etc. To the extent fossil fuel companies are more likely to issue green bonds, and these are the bonds that the California insurers hold in the post-period, then we would be undercounting the effect the disclosure has on insurers' fossil fuel holdings. As an extreme example, if insurers divested of all fossil fuel bonds except for green bonds, then the remaining amount of investment we observe would all be green bonds. To the extent that such bonds are considered a positive in the sustainability framework, we would be misrepresenting them as continued negative fossil fuel investment. Unfortunately, we are unaware of any data source that would allow us to identify green bonds.

3.1.2. Parallel trends assumption

One of the major assumptions of the difference-in-difference research design is that, in the absence of treatment, the treatment and control groups would have followed parallel trends over time. While we cannot observe the trends that would have existed absent treatment for the California firms (i.e. a post rule counter factual), we can examine the trends in the years prior to treatment to determine if they were similar. If ownership trends are similar between the California-licensed insurers and all others, this provides some assurance that, absent varying treatment, the two groups would have followed similar future investment trends. We test this by substituting *Post* in the main specification with quarter-year indicator variables with the base quarter set on September 30, 2016, which is the quarter just prior to the enactment of the California Rule. The results are presented in Table 4. $Q = 1$ refers to observations on 12/31/2016, $Q = 2$ are observations on 3/31/2017, and so on. The coefficient on the interaction term of *Treat* with the quarter-year dummy shows there is no statistical difference between the two groups prior to the California Rule. At $Q = 1$, the coefficient is negative (coef. = −0.006, t-stat = −1.22) and starting at $Q = 2$ the negative coefficient is statistically significant (coef. = −0.008, t-stat = −1.66). The negative and significant coefficient remains for the remainder of the sample period.

Fig. 3 plots the coefficient on the interaction term over the sample period. The disclosing and non-disclosing insurers are on similar trends prior to the California Rule at which point they diverge. The disclosing insurers show a greater reduction in their fossil fuel investments in the post-period. Overall, the results provide evidence in support of the parallel trends assumption.[33]

3.2. Why do some firms divest?

3.2.1. Regulatory pressure

Having shown that insurers divest their fossil fuel securities, we next explore *why* they divest by examining cross-sectional factors within the set of treated (Californian) insurers. First, insurers may feel direct regulatory pressure to divest. The commissioner has considerable influence on the insurers via approving rate increases and potentially impacting other business opportunities. The commissioner may directly threaten (or imply) to hold back rate increases or make the process more burdensome unless firms reduce their fossil fuel investments. Alternatively, the insurers may plausibly fear such actions even without direct or implied threats. In a similar setting, Barber et al. (2021) find that regulatory pressure encourages investment in impact funds.[34] Overt or assumed regulatory pressure is particularly salient in this setting as the commissioner publicly stated they would like divestment at the time they announced the disclosure requirement.

To measure this regulatory pressure, we identify insurers with strong ties to California proxied for with the following variables: (i) *California Premiums (%)* – the fraction of an insurer's premiums generated in California, and (ii) *California HQs* – an indicator variable equal to one if the insurer is headquartered in the state and zero otherwise. While these proxies are not perfect, we assume that those insurers that are more dependent on California for economic success or are headquartered in the state are more susceptible to pressure from the regulator and, thus, more likely to appease the regulator and divest their fossil fuel investments as a result.[35]

The results of the regulatory pressure test are presented in columns (1) and (2) of Table 5. Our sample in this table is limited to California licensed insurers, as we are studying the impact on treated firms. In column (1), we look at variation in *California Premiums (%)*. The coefficient on the interaction term is positive but insignificant, suggesting there is no difference in fossil fuel divestment between those insurers with a high percentage of premiums from California customers vs a lower percentage. In column (2), we look at *California HQs,* and again, the coefficient on the interaction term is positive but insignificant, suggesting no difference between insurers headquartered in California or outside the state. Our results provide no evidence that firms that face greater regulatory pressure are more likely to divest fossil fuel securities, suggesting the declines documented in prior tests are not due to underlying regulatory actions or fears.

3.2.2. External stakeholder pressure

Variance in disclosure may also be due to variance in actual or perceived scrutiny from external stakeholders. Increasingly,

[33] Both Fig. 3 and Table 4 show there is significant quarter-to-quarter variation in fossil fuel ownership. The start of the California rule on 12/31/2016 elicits a sizeable decline in that quarter. As the quarters progress in the post-period, the coefficients become statistically significant. This shows the short-window effect around the start of the rule.

[34] These are investments made to "generate positive, measurable social and environmental impact alongside a financial return" (Barber et al., 2021).

[35] A counterargument is that the insurance commissioner could be more lenient towards a firm headquartered in California or conducting most of their business within the state. However, we consider this unlikely for several reasons. First, the commissioner has substantial authority over the in-state insurers. As we mentioned, the commissioner sets rates, and the California insurers fear upsetting the commissioner. As one insurance industry observer said at the time of the California Rule's announcement, "[I]nsurers are in something of a bind when it comes to rebutting any insurance commissioner. The commissioner has so much authority over insurers that they rightly fear retribution when it comes to, for example, rate hike requests. So, while the commissioner and his supporters say the divestment request is voluntary, it's easy to see why insurance companies might believe there is some duress involved" (Greenhut, 2016). Second, at the time, the commissioner had future political aspirations. He was incentivized to be tough on California insurers, especially in his last term as commissioner, to secure a "political gain" to help with his future campaign. Lastly, the commissioner's position is designed to have much autonomy. The voters elect the commissioner. There is no legislative session to approve rules. All these factors make it less likely that the commissioner was lenient to the in-state insurers. Also, our results show no association between the regulatory pressure proxies and divestment. Even if the commissioner was more lenient to California insurers and stricter to non-California insurers this does not affect their fossil fuel investment.

Table 4
Effect of mandatory disclosure on fossil fuel investments over time.

Dep. Var =	*Fossil Fuel Securities/Tot Bonds*		
	coef.	t-stat	sign.
	(1)	(2)	(3)
Treat × *Q* = *-10*	0.001	(0.24)	–
Treat × *Q* = *-9*	0.001	(0.28)	–
Treat × *Q* = *-8*	0.000	(-0.01)	–
Treat × *Q* = *-7*	0.005	(0.83)	–
Treat × *Q* = *-6*	0.004	(0.80)	–
Treat × *Q* = *-5*	0.003	(0.46)	–
Treat × *Q* = *-4*	0.001	(0.19)	–
Treat × *Q* = *-3*	0.002	(0.37)	–
Treat × *Q* = *-2*	0.001	(0.44)	–
Treat × *Q* = *-1*	0.000	(-0.34)	–
Treat × *Q* = *1 (12/31/2016)*	−0.006	(-1.22)	–
Treat × *Q* = *2*	−0.008	(-1.66)	*
Treat × *Q* = *3*	−0.008	(-1.76)	*
Treat × *Q* = *4*	−0.011	(-2.13)	**
Treat × *Q* = *5*	−0.012	(-2.32)	**
Treat × *Q* = *6*	−0.012	(-2.41)	**
Treat × *Q* = *7*	−0.012	(-2.24)	**
Treat × *Q* = *8*	−0.012	(-2.47)	**
Treat × *Q* = *9*	−0.015	(-2.52)	**
Treat × *Q* = *10*	−0.014	(-2.47)	**
Treat × *Q* = *11*	−0.014	(-2.44)	**
Treat × *Q* = *12*	−0.013	(-2.44)	**
Treat × *Q* = *13*	−0.012	(-1.94)	*
Controls	Yes		
Firm FE	Yes		
N	43,820		
R-squared	0.818		

Notes. This table displays the dynamic effects of the California Rule on insurers' fossil fuel investments. The dependent variable *Fossil Fuel Securities/Tot Bonds* is the par value of fossil fuel investments divided by the total par value of the firm's bond portfolio. We interact *Treat* × *Quarter*, where *Treat* is equal to one for California-licensed insurers and zero otherwise. The base quarter is set as September 30, 2016, the quarter just prior to the enactment of the California Rule. *Q* = *1* refers to observations at 12/31/2016, *Q* = *2* are observations at 3/31/2017 and so on. Columns (1) through (3) display the estimated coefficient, t-statistic and level of significance respectively.

stakeholders view fossil fuel investments as a negative attribute (Hong et al., 2020). The disclosures brought the insurer's fossil fuel investments to the attention of environmental activists (Messervy, 2016) as well as the broader public, including shareholders (Reuters, 2016; Bloomberg, 2016). These groups could voice their concerns and discipline insurers that invest in fossil fuel companies. Even a perceived threat from these groups could motivate insurers to divest (Fineman and Clarke, 1996).

To measure external stakeholder pressure, we look at the influence of two groups on insurers - environmental activists and public shareholders. The presence or threat of environmental activists can result in negative publicity that can hurt a company's sales or reputation (Lewis et al., 2017). Firms that have already faced pressure from these groups may view the disclosure of investment in fossil fuels as creating another opportunity to attract activist attention and focus on the firm's business practices. We create an indicator variable*, Environmental Activists,* equal to one if the insurer has been the target of an activist group prior to the California Rule. We specifically look at campaigns by well-known organizations, including Greenpeace, Unfriend Coal, Insure Our Future, and Public Citizen.

Shareholders may also pressure the insurers to divest to avoid actual or perceived reputational harm (Hombach and Sellhorn, 2019). We use two measures to capture potential shareholder pressure. First, we create *Public,* defined as an indicator variable equal to one if the insurer is publicly traded and zero otherwise. Second, we create *Market value,* which is an indicator variable coded as one if the firm is both publicly traded and has a market value greater than the median publicly traded firm and zero otherwise. Finally, it is also possible that larger firms simply attract more stakeholder interest in general. We create the indicator variable *Assets* coded as one if the insurer's total assets are greater than the median and zero otherwise. While it is highly correlated to both *Public* and *Market Value*, it does allow very large private firms to be considered as targets as well. Obviously, the three market and size-based variables are not independent but rather provide different constructs for identifying firms with high visibility to a large degree of external stakeholders.

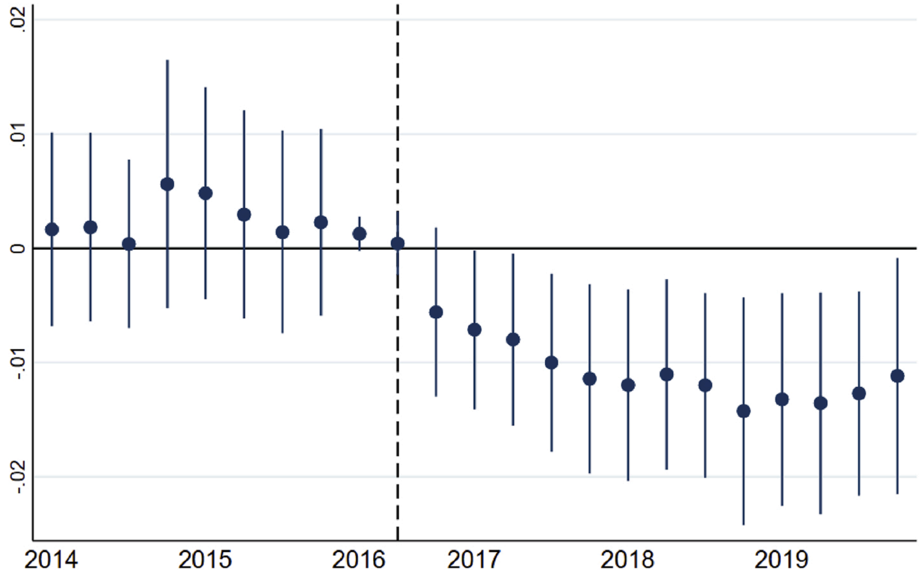

Fig. 3. Effect of Disclosure on Fossil Fuel Holdings Over Time
Notes. This figure displays the dynamic effects of the California Rule on insurers' fossil fuel investments. We interact *Treat* × *Quarter-Year*, where *Treat* is equal to one for California-licensed insurers and zero otherwise. The base quarter is 9/30/2016. We plot the estimated coefficient along with the 90 % confidence intervals.

Table 5
Economic mechanisms.

Dep. Var =	*Fossil Fuel Securities/Tot Bonds*					
Classification =	Regulatory Pressure		External Stakeholder Pressure			
Partitioning Var =	*California Premiums (%)*	*California HQ*	*Environmental Activists*	*Public*	*Assets (1,0)*	*Market Value*
	(1)	(2)	(3)	(4)	(5)	(6)
Partitioning Var × Post	0.0138	0.00882	−0.0212*	−0.0234*	−0.0186**	−0.0431*
	(1.58)	(1.22)	(-1.79)	(-1.66)	(-2.12)	(-1.68)
Gross Premiums	0.000900	0.00137	0.00116	0.00130	0.00101	0.00121
	(0.43)	(0.67)	(0.58)	(0.66)	(0.48)	(0.31)
Investment Income (%)	−0.00235	−0.00310	0.00000284	−0.00155	−0.00239	−0.000614
	(-0.30)	(-0.38)	(0.00)	(-0.19)	(-0.29)	(-0.03)
Constant	0.0757**	0.0706**	0.0776**	0.0754**	0.0818**	0.124*
	(2.13)	(2.01)	(2.22)	(2.17)	(2.23)	(1.83)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Quarter-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
N	7500	7500	7500	7500	7500	2204
R-squared	0.821	0.821	0.822	0.823	0.822	0.819

Notes. This table examines *why* the disclosing insurers reduce their fossil fuel investments. We restrict the sample to all California-licensed insurers and run cross-sectional tests with different partitioning variables. *California Premiums (%)* is an indicator variable equal to one if the insurer's percentage of premiums that come from California is greater than the median value and zero otherwise. *California HQs* is equal to one for those insurers headquartered in California and zero otherwise. *Environmental Activists* is equal to one if the insurer was a target of an activist investor prior to 2016. *Public* denotes whether the insurer is publicly-traded or not. *Assets (1,0)* is an indicator variable equal to one for insurers with total asset value above the median and zero otherwise. *Market Value* is an indicator variable equal to one for publicly traded firms with a market valuation above the median and zero otherwise. *Post* is an indicator variable equal to one for the period after the announcement of the California Rule (from 12/31/2016 onwards) and zero for the prior period. Variable definitions are in Appendix A. Standard errors are clustered at the firm level. T-stats are displayed in parentheses. *, **, and *** indicate significance levels of 10 %, 5 % and 1 %, respectively.

The tests of the external stakeholder pressure are presented in columns (3) through (6) of Table 5. Focusing on the interaction term, column (3)'s negative coefficient shows that firms with prior activist targeting are more likely to divest, consistent with the disclosures attracting more attention or being used by these groups. Turning to the measures of investors, we find that both indicate firms with more investor pressure are more likely to divest fossil fuel holdings, again consistent with the disclosures impact being related to the power of external stakeholders. Finally, we find a positive relation between size and divestment, also indicating that firms with higher visibility and potentially more stakeholders are more likely to divest following disclosure. We test whether the coefficients across the regulatory pressure and external stakeholder pressure partitioning variables are statistically different. Using a Wald Chi-Squared test, we find that the differences in coefficients are statistically significant. This result further provides supporting evidence that insurers respond more strongly to external stakeholder pressure than to regulatory pressure.

3.3. Fossil fuel holdings once the disclosure stops

The California Rule was enacted by then-commissioner Dave Jones and stopped when his term ended in 2019. This creates a unique feature that allows us to observe how the firms' behaviors change after the mandatory disclosure ceases. The results are presented in Table 6. We follow the main specification and restrict the sample to the years 2018–2021. *Treat x Post (= 2020)* is an indicator variable equal to one for California-licensed insurers in the post-period after the disclosure regulation ends (i.e., 2020–2021). Columns (1) and (2) present the results with firm and quarter-year fixed effects, while columns (3) and (4) present the results with firm and state-year fixed effects. Columns (2) and (4) include control variables, including the gross premiums, assets, and percentage of income from investment gains or losses. In all four columns, the coefficient on the interaction term is insignificant indicating there is no statistically significant change in fossil fuel investments among the disclosing insurers once the California Rule ends. In other words, the insurers do not revert to their pre-policy holdings of fossil fuel investments. This suggests that the effect of the mandatory disclosure was not fleeting. However, the lack of significant changes also indicates firms did not further decrease their holdings after the disclosure treatment was removed, indicating no further divestment occurred.[36]

3.4. Effect of the California rule on firm's investment policies

Having documented that the investment changes were sustained after the rule was rescinded by a new commissioner, we explore whether changes in investment policy are related to this investment stickiness. In the following section we use insurers responses to an investment survey to identify and examine investment policies.

3.4.1. Climate risk survey

Beginning in 2010, several states have required insurance companies licensed in the state to respond to a nine-question climate risk survey called the NAIC Climate Risk Survey.[37] Since the surveys existed both pre and post the California rule, we can examine if the rule is associate with changes in insurers investment strategies and risk management. We focus on their responses to three questions: (i) does the company have a climate change policy with respect to risk management and investment management, (ii) has the company considered the impact of climate change on its investment portfolio, and (iii) has [the insurer] altered its investment strategy in response to these considerations. The insurers are asked to provide a "yes" or "no" answer and to explain their responses. See Appendix C for an example of the survey.

Table 7 presents the effect of mandatory disclosure on insurers' investment policies. For these tests we follow the main specification as outlined in Eq. (1). The dependent variable is equal to one if the insurer answers "yes" to the question and zero if they answer "no." *Treat* × *Post* is a binary variable that equals one for California licensed insurers after the announcement of the California Rule (i.e., 12/31/2016 onwards). Column (1) presents the results from the question: does the company have a climate change policy with respect to risk management and investment management? The coefficient on the interaction term is positive and significant (coef. = 0.0485, t-stat = 2.56), suggesting the onset of the California Rule encourages California based insurers to have a climate change policy. Column (2) presents the results from the question: has the company considered the impact of climate change on its investment portfolio? Column (3) presents the results from the follow-up question: has [the insurer] altered its investment strategy in response to these considerations? Across both, the coefficient on the interaction term is positive and significant indicating the disclosing firms are more likely to consider the effect of climate change on their investment portfolio and more likely to alter their investment strategy after disclosure became required.

The results show that the California Rule prompts treated firms to consider climate change in their portfolios, adjust their investment strategy, and adopt a risk management policy. Together with the results in the previous section, this suggests the California

[36] This finding also provides further support that the disclosure rule caused the observed changes in fossil fuel holdings for the California firms. Any alternative explanation would need to i) only impact California firms ii) start at the same time as the disclosure rule and iii) stop at the same time as the disclosure rule.

[37] As of 2021, 14 states require the disclosure from over 1400 insurance companies. Those responding account for over 80 % of the premium volume in the U.S. The states include California, Connecticut, Delaware, Maine, Maryland, Massachusetts, Minnesota, New Mexico, New York, Oregon, Pennsylvania, Rhode Island, Vermont and Washington along with the District of Columbia. In April 2022, the NAIC adopted a new standard for insurance companies to report their climate-related risks which follows the guidance of the Task Force on Climate-Related Financial Disclosures (TCFD) (NAIC, 2023).

Table 6
Investment when the disclosure regulation ceases.

Dep. Var =	*Fossil Fuel Securities/Tot Bonds*			
	(1)	(2)	(3)	(4)
Treat × Post (= 2020)	−0.00192	−0.00187	−0.000922	−0.000884
	(-0.95)	(-0.92)	(-0.47)	(-0.45)
Gross Premiums		0.00110		0.00116
		(1.12)		(1.23)
Assets		0.00103		0.000755
		(0.92)		(0.71)
Investment Income (%)		0.000231		−0.0000838
		(0.38)		(-0.14)
Constant	0.0586***	0.0210	0.0584***	0.0247
	(113.92)	(1.35)	(221.22)	(1.62)
Firm FE	Yes	Yes	Yes	Yes
Quarter-Year FE	Yes	Yes	No	No
State-Year FE	No	No	Yes	Yes
N	27,624	27,624	27,624	27,624
R-squared	0.885	0.885	0.891	0.891

Notes. This table examines changes to insurers' fossil fuel holdings before and after the disclosure regulation ceases. The sample period is from 2018 to 2021. *Treat x Post (= 2020)* is an indicator variable equal to one for California-licensed insurers in the post-period after the disclosure regulation ends (2020–2021). *Fossil Fuel Securities/Tot Bonds* is the par value of fossil fuel investments divided by the total par value of the firm's bond portfolio. We include firm and quarter-year fixed effects in columns (1) and (2) and firm and state-year fixed effects in columns (3) and (4). Variable definitions are in Appendix A. Standard errors are clustered at the state-year level. T-stats are displayed in parentheses. *, **, and *** indicate significance levels of 10 %, 5 % and 1 %, respectively.

Rule resulted in both portfolio changes and policy changes. The policy changes, in particular, may be one reason why the disclosing insurers do not immediately buy back the fossil fuel securities when the disclosure mandate ends.[38]

We further use the survey responses to better understand the motivations for divestment.[39] We read through the responses and found several explanations from the insurers. First, they specifically mention the reputational harm of owning fossil fuel holdings.[40] Second, insurers say the change in investment policies complements larger sustainability pledges.[41] Finally, some insurers indicate they want to be known for their environmental stewardship.[42]

[38] Another reason why the insurers may not immediately buy back the divested fossil fuel securities could be due to reporting changes to the NAIC survey in the post-policy period. Starting in 2022, the survey changed to become compliant with the Task Force on Climate-related Financial Disclosures (TCFD). Some of the California insurers may have decided to not buy back the fossil fuel securities knowing the more rigorous reporting were soon to be implemented. There are also transactional costs the insurers would incur from selling their holdings to buy back the fossil fuel securities that may dissuade them from adjusting their portfolios. Finally, there may be other reputation concerns for the insurer and the fund managers if it becomes widely known they bought fossil fuel securities once the disclosure regulation ceased. It is difficult to quantify these costs but they seem non-trivial. For example, changing formal investment policies can be time- and labor-intensive. Revising an investment policy statement involves careful deliberation and consensus building among a variety of parties in and outside the firm which can be a lengthy process (Goldman Sachs, 2023). Not only do new policies need to be written and passed but internal investment processes will change as a result.

[39] In addition to reviewing responses to the survey, we examined other channels insurers could use to explain their decisions to divest or their response to the California rule. We find no evidence of discussion in press releases issued in the year leading up to the California Rule nor during the sample period. Similarly, a review of the quarterly conference call transcripts for the publicly-traded California insurers finds no specific mentions of the California rule.

[40] As one insurer said, "The Working Group's initial risk assessment showed there are only minor risks to Aegon's business from climate change; our main risks are reputational. We will need to continue to monitor and manage these risks as part of our overall approach to risk management."

[41] As explained by this insurer: "Climate change issues, other environmental and legal scenarios are part of that consideration in analyzing industries and issuers. In fact, our investment manager is a signatory to the United Nations Principles for Responsible Investments (UNPRI) group and climate change is one focus of the group."

[42] For example, Kaiser Permanente said it is "committed to be a model for environmental stewardship, not only in the health care space, but as a local, state, national and global corporate citizen and to partner with others to follow an ambitious sustainability path that will help reduce global warming and prevent irrevocable damage."

Table 7
Effect of mandatory disclosure on investment policies.

Dep. Var =	*Risk & Investment Mgmt Policy*	*Portfolio*	*Strategy*
	(1)	(2)	(3)
Treat × Post	0.0485**	0.0520***	0.0825***
	(2.56)	(2.96)	(3.04)
Gross Premiums	0.00383	−0.00509	0.00224
	(0.50)	(-1.07)	(0.25)
Assets	0.0192	0.000733	−0.0223*
	(1.20)	(0.07)	(-1.65)
Investment Income (%)	−0.0113	0.0224	0.0626
	(-0.27)	(0.72)	(0.99)
Constant	0.0947	0.869***	0.691**
	(0.37)	(4.41)	(2.49)
Firm FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
N	4623	4623	4623
R-squared	0.797	0.746	0.702

Notes. This table presents the effect of the California Rule on insurers' responses to the NAIC Climate Risk survey. The sample consists of annual observations of insurers with available responses to the survey. The dependent variable is equal to one if the insurer answers "yes" and zero if the insurer answers "no" to the following three questions. For column (1) the question is "does the company have a climate change policy with respect to risk management and investment management?" Column (2) asks "has the company considered the impact of climate change on its investment portfolio?" And column (3) asks "has [the insurer] altered its investment strategy in response to these considerations?" *Treat × Post* is a binary variable that equals one for California licensed insurers after the announcement of the California Rule (i.e. 12/31/2016 to 2019). All columns use firm and year fixed effects and a host of control variables. Variable definitions are in Appendix A. Standard errors are clustered at the state-year level. T-stats are displayed in parentheses. *, **, and *** indicate significance levels of 10 %, 5 % and 1 %, respectively.

3.4.2. Cross-sectional tests

We next examine the relationship between divestment intensity and the stated changes in investment policies. The results are presented in Table 8. For these tests, we again restrict the sample to only California-licensed insurers. The dependent variable is *Fossil Fuel Securities/Tot Bonds*. *Policy Change* is an indicator variable equal to one if the insurer adopts a risk and investment management policy after the onset of the California Rule. *Post* is a binary variable equal to one for the period after the announcement of the California Rule (from 12/31/2016 onwards) and zero for observations in the prior period. We follow the main specification of Eq. (1) and include firm and quarter-year fixed effects and cluster standard errors at the firm level. Column (1) presents the results with only firm and quarter-year fixed effects. Column (2) adds firm-specific control variables, and column (3) includes changes in commodity prices. Across all three columns, the coefficient on the interaction term is negative and significant. This shows that insurers that adopt a risk and investment management policy with respect to climate change have a greater divestment intensity during that period.

In untabulated results, we divide the sample into three mutually exclusive categories – those that previously had policies, those that switched to having policies during the mandated disclosure policy, and those that answered no to the question of whether they have an investment policy to consider fossil fuels. While we find a statistically significant decline in fossil fuel investment for all firms with an investment policy, those that adopt a policy during the disclosure period have a −0.0392 reduction, while those who always had a policy have a −0.0066 reduction. In contrast the divesture coefficient on those with no policy is smaller and statistically insignificant. Similar to our findings on investment decisions, untabulated analyses also show that firms do not change their investment policies when the disclosure rule is no longer in effect. Overall, these findings support the conclusion that the mandatory disclosure created fundamental changes in many firms' investment policies and decisions.

4. Additional results

4.1. Is the divestment real divestment?

While some firms appear to have decreased fossil fuel investment we consider two strategic actions: window dressing and transferring firms to a related subsidiary - that would indicate divestment, but economically not change firm holdings. Firms window could window dress by making sales at year end and repurchasing the same assets soon after the beginning of the new year. Window dressing

Table 8
Investment policies – cross-sectional tests.

Dep. Var =	*Fossil Fuel Securities/Tot Bonds*		
Sample =	CA-licensed Insurers		
	(1)	(2)	(3)
Policy Change × Post	−0.0350**	−0.0352**	−0.0352**
	(-2.04)	(-2.05)	(-2.06)
Gross Premiums		0.00652*	0.00651*
		(1.92)	(1.92)
Total Assets		−0.00682	−0.00681
		(-1.51)	(-1.51)
Investment Income (%)		−0.00708	−0.00708
		(-0.93)	(-0.94)
Post			−0.000458
			(-0.20)
Coal Prices			−0.0264
			(-0.89)
Natural Gas Prices			0.0128***
			(3.15)
Oil Prices			−0.0123***
			(-4.16)
Constant	0.0990***	0.107**	0.107**
	(42.49)	(2.44)	(2.43)
Firm FE	Yes	Yes	Yes
Quarter-Year FE	Yes	Yes	No
Year FE	No	No	Yes
N	7500	7500	7500
R-squared	0.825	0.826	0.825

Notes. This table examines the change in fossil fuel investments for those California-licensed insurers that do or do not adopt a risk and investment management policy. The sample consists of quarterly observations. The dependent variable is *Fossil Fuel Securities/Tot Bonds*. *Policy Change* is an indicator variable equal to one if the insurer changes their answer (from "No to "Yes") to the question, does the company have a climate change policy with respect to risk management and investment management?" *Post* is a binary variable equal to one for the period after the announcement of the California Rule (from 12/31/2016 onwards) and zero for the prior period. We include firm and quarter-year fixed effects. Variable definitions are in Appendix A. Standard errors are clustered at the firm level. T-stats are displayed in parentheses. *, **, and *** indicate significance levels of 10 %, 5 % and 1 %, respectively.

has been shown among other investors including banks (Allen and Saunders, 1992) and mutual funds (Agarwal et al., 2014). Similarly, the large number of consolidations in the field (Insurance Information Institute, 2021), creates an opportunity to move the investments from a California subsidiary to one in another state which is not required to disclose. At the consolidated level, the parent company would still have the same holdings and avoid any costs from an abrupt sale. We test for the presence of window dressing and sales to related insurers in Panels A and B of Table 9.

Panel A presents the results of the window dressing test. Using the end-of-year portfolio and the transactions data, we reconstruct the information on the holdings of each insurer at the monthly level. We restrict the sample to observations three months before and three months after the end of 2016, 2017, and 2018, respectively. We follow the main specification as outlined in Eq. (1) with the difference that *Post* is equal to one for observations in the first quarter of each year and zero for observations in the 4th quarter of the previous year. If window dressing occurs, we would observe a positive and significant coefficient on the interaction term, suggesting the disclosing insurers increase their fossil fuel holdings immediately at the start of the next year. *Treat* × *Post* is a binary variable that equals one for California licensed insurers after the announcement of the California Rule (i.e., 12/31/2016 onwards). Columns (1), (2), and (3) restrict the sample to monthly holdings at the end of 2016, 2017, and 2018, respectively, while column (4) combines all three years. If insurers engage in window dressing the coefficient would be positive, indicating they increase their fossil fuel holdings after the annual reporting date has passed. However, we find no statistically significant difference in fossil fuel holdings from the end of one

Table 9

PANEL A: Window Dressing at the end of the year

Dep. Var =	*Fossil Fuel Securities/Tot Bonds*			
Sample =	Q4 2016 to Q1 2017	Q4 2017 to Q1 2018	Q4 2018 to Q1 2019	Pooled
	(1)	(2)	(3)	(4)
Treat × Post (= Q1)	−0.00287	−0.000288	−0.000788	−0.00134
	(-1.34)	(-0.46)	(-0.91)	(-0.83)
Gross Premiums	0.0000382	−0.000376	0.00139*	−0.00102
	(0.14)	(-1.08)	(1.68)	(-1.32)
Total Assets	0.000655	0.000270	−0.00139	0.0000587
	(0.47)	(0.55)	(-1.50)	(0.08)
Investment Income (%)	0.00201	0.00359***	0.00131*	0.00267**
	(0.50)	(2.63)	(1.91)	(2.40)
Constant	0.0574**	0.0651***	0.0594***	0.0805***
	(2.35)	(10.98)	(6.56)	(7.27)
Firm FE	Yes	Yes	Yes	Yes
Quarter-Year FE	Yes	Yes	Yes	Yes
N	11,121	10,854	10,569	32,544
R-squared	0.968	0.990	0.978	0.921

Notes. Panel A presents the results of the window dressing test. We use monthly observations and restrict the sample to observations three months before and three months after the end of 2016, 2017 and 2018, respectively. We follow the main specification with the difference that *Post* is equal to one for observations in the first quarter of each year and zero if the observation is in the 4th quarter of the previous year. The dependent variable *Fossil Fuel Securities/Tot Bonds* is the par value of fossil fuel investments divided by the total par value of the firm's bond portfolio. Columns (1), (2) and (3) restrict the sample to monthly holdings at the end of 2016, 2017 and 2018 respectively while column (4) combines all three years. We include firm and quarter-year fixed effects. Variable definitions are in Appendix A. Standard errors are clustered at the state-year level. T-stats are displayed in parentheses. *, **, and *** indicate significance levels of 10 %, 5 % and 1 %, respectively.

PANEL B: Related Party Transactions

Dep. Var =	*Sold to Related Insurer*		
	(1)	(2)	(3)
Treat × Post	0.0209	0.0202	0.0202
	(0.56)	(0.53)	(0.53)
Gross Premiums - Seller		−0.000940	0.00161
		(-0.04)	(0.04)
Assets - Seller		0.0147	0.0498
		(0.44)	(1.23)
Investment Income (%) - Seller		−0.00734	0.0248
		(-0.21)	(0.38)
Gross Premiums - Buyer			−0.00337
			(-0.08)
Assets - Buyer			−0.0409
			(-0.95)
Investment Income (%) - Buyer			−0.0338
			(-0.59)
Post	0.619***	0.619***	0.619***
	(6.45)	(6.42)	(6.42)

(*continued on next page*)

Table 9 (*continued*)

PANEL B: Related Party Transactions			
Dep. Var =	*Sold to Related Insurer*		
	(1)	(2)	(3)
Constant	0.0406	−0.206	−0.0874
	(0.80)	(-0.49)	(-0.20)
Firm FE	Yes	Yes	Yes
Quarter-Year FE	Yes	Yes	Yes
N	6455	6455	6455
R-squared	0.576	0.576	0.576

Notes. Panel B presents the results of the related party transactions test. The dependent variable *Sold to Related Insurer* is equal to one if the insurer sells a fossil fuel investment to an insurer with the same parent company and zero otherwise. A security is considered sold to another insurer if they buy the security (with same 9-digit CUSIP) within one week of the selling insurer disposing of the security. *Treat* is equal to one for California-licensed insurers and zero otherwise. *Post* is equal to one for observations on 12/31/2016 or thereafter and zero otherwise. Column (1) presents the results with only firm and quarter-year fixed effects. Column (2) adds seller insurer control variables and column (3) adds buyer insurer control variables. Variable definitions are in Appendix A. Standard errors are clustered at the state-year level. T-stats are displayed in parentheses. *, **, and *** indicate significance levels of 10 %, 5 % and 1 %, respectively.

year to the beginning of the next.

Panel B presents the results of the related party transactions test. The dependent variable is equal to one if the insurer sells a fossil fuel investment to an insurer with the same parent company and zero otherwise. *Treat* is equal to one for California-licensed insurers and zero otherwise. *Post* is equal to one for observations on 12/31/2016 or thereafter and zero otherwise. Column (1) presents the results with only firm and quarter-year fixed effects. Column (2) adds seller insurer control variables, and column (3) adds buyer insurer control variables. The coefficient on the interaction term across all three columns is insignificant suggesting the insurers do not sell their fossil fuel securities to related insurers. Both the window dressing and related party tests show that the divestment of fossil fuel securities are *real* divestments instead of just the *appearance of* divestment.[43]

4.2. *What is the effect on the fossil fuel companies?*

Given that we find the disclosing insurers reduce their fossil fuel investments with the onset of the California Rule, the natural follow-up question is what effect, if any, does this have on the fossil fuel companies? One possible impact is an increase in the cost of borrowing. An increase would be a direct cost to the firm, and if substantial enough, could pressure fossil fuel companies to reduce their greenhouse gas emissions. We examine the change in the yield for newly issued bonds by fossil fuel companies before and with the onset of the California Rule. We focus on the initial yield because it determines the firm's borrowing cost. *Ex-ante* it is hard to predict whether there would be price pressure from the regulation. If the rule had a significant effect and enough investors stopped participating in new bond offerings, we should see higher borrowing costs. On the other hand, the California insurers are a relatively small portion of the universe of potential investors, and other investors may have stepped up to purchase the offering bonds, muting any price effect. In untabulated results, we find no statistically significant change in the bond yield with the onset of the disclosure. While the yield may not change, the fossil fuel companies may reduce the size of their bond offerings. We test this and again find no statistically significant change. The evidence suggests that by only impacting some insurers, the California Rule was too limited to influence security prices, borrowing costs and bond offering amounts.[44]

4.3. *What are the costs the insurers bear to divest?*

At the time of the California Rule's announcement, many feared the mandatory disclosure and divestment would hurt the insurers' returns and weaken their financial position as the insurers would sell bonds at unfavorable prices and forego higher returns in the future (Greenhut, 2016). Others argued that any price impact would be negligible as the fossil fuel securities represented a relatively small percentage of their total bond portfolio, and the insurers had considerable time to make the divestment.

[43] Some critics of the California Rule argued that California insures would simply sell to other insurers outside of California and thus there would be no impact on the fossil fuel companies (Greenhut, 2016). To examine this, we track whether non-California insurers purchase the same security being divested by a Californian company within 7 days of divestment. Our results, in the online appendix, show that approximately 90 % of the sales are to non-insurance companies. We also find the insurers replace the divested fossil fuel securities with securities that have higher credit quality, lower bid/ask spreads, lower yield to maturities, and higher durations. These securities are issued by companies in regulated industries like telephone communications, banks and pharmaceuticals.

[44] It would be ideal to be able to directly test for changes in emissions among the issuing firms. However, few firms publicly report this information over our sample period and there was no standardized way to report emissions. It would be difficult to compare across firms or even across the same firm at different times as reporting emissions, and the technology to track emissions changed over time (Cohen et al., 2025).

In this section, we examine the impact of the disclosure on investment returns. The results presented in Table 10 suggest that insurers suffer lower returns with the onset of the disclosure and subsequent divestments. To calculate the returns, we sum the unrealized gains or losses and the interest payments received and accrued for the insurer's total bond portfolio each year. We scale the amount by the size of the bond portfolio to create *Tot Return*. We follow the prior tests and examine the change in returns between the disclosing and non-disclosing insurers before and after the announcement of the California Rule. Column (1) shows the coefficient on the interaction term is negative and significant (coef. = −0.00101, t-stat = −2.24), suggesting the disclosing insurers experience lower returns. If divestment causes poor performance, the lower returns should be concentrated in those insurers who divest the most from the pre-to the post-period. We partition the California-licensed insurers based on the change in their fossil fuel holdings in those two periods. In Columns (2) and (3), we partition the sample with *High Divestment,* denoting disclosing firms that decreased their fossil fuel holdings at a rate higher than the median and *Low Divestment* for those below the median. The reduction is concentrated in the *High* group, which suggests that those insurers who divest the most experience the worst subsequent returns. Those in the *High* group experienced 0.2 % lower returns in the post-period. With an average total bond portfolio of approximately $500 million, this means they incur $1 million in additional costs due to divestment. We present the results of a Wald Chi-Squared test which shows that the coefficients from the *High* and *Low* groups are statistically different from one another (diff. = −0.0018, p-value = 0.068). The results support the concerns mentioned at the time of the rule's announcement. Divestment did come at a cost to the disclosing insurer.

As with most accounting papers, we cannot provide a global social welfare test. To do so, we would have to model the costs and benefits to customers, shareholders, lenders, employees, and the impact on the population as a whole. However, by combining the evidence in sections 4.2 and 4.3, we can see that fossil fuel companies face no higher borrowing costs from this change and thus have little incentive to change, while insurers generate lower returns on their portfolio. That loss in return must be borne by some stakeholders – customers through higher prices or a higher risk of default by the insurer and shareholders via a lower return, as just two examples. On balance, the rule does not appear to have provided much incentive for issuers to change, but it does appear to have imposed some costs on stakeholders. Again, this conclusion should be viewed with the caveat that we cannot provide a comprehensive social welfare model.

4.4. *Sensitivity tests*

In this section, we explore a variety of alternative specifications, samples, and proxies for our main variables. The results are presented in Table 11.

4.4.1. *Alternative specifications*

Panel A presents the coefficient and t-stat for the main coefficient (*Treat* × *Post*) based on various model specifications. In row (1) with no firm or quarter-year fixed effects. In row (2), we replace firm fixed effects with parent company fixed effects to account for unobservable time-invariant characteristics at the parent company level. Parent companies, or holding companies, often own several insurance companies (Insurance Information Institute, 2021). There may be similar cultures, investment strategies, and managers across the insurers within the same parent company. In row (3), we cluster the standard errors by quarter-year. In row (4), we account for the possibility that the effect of covariates varies across treated and control firms by repeating the baseline analyses and interacting each control variable with *Treat*. Similarly, in row (5), to account for the possibility that the effect of covariates varies across the pre- and post-period, we repeat the baseline analyses but interact each control variable with *Post*. Finally, row (6) uses one-to-one propensity score matching without replacement matching on four characteristics: gross premiums, total assets, the value of the bond portfolio, and the average value of the fossil fuel investments held by the insurer in the pre-period. The coefficient of interest remains negative and significant across all the different model specifications.

4.4.2. *Alternative samples*

Panel B presents the results with different samples. In row (7), we exclude Californian insurers below $100M in nationwide premiums. In row (8), we restrict the sample to insurers domiciled in states that require the NAIC Climate Risk Survey. The effect documented in this paper may be due to insurers responding to the survey and not the specific disclosure regulation. In row (9), we restrict the sample to insurers domiciled in states with carbon cap-and-trade programs at the time of the California Rule to ensure changes in carbon prices do not drive the results. [45] In row (10), we use end-of-year rather than quarterly holdings. In row (11), we restrict the sample to Californian insurers with observations immediately above or below the $100 million premium threshold. The coefficient of interest remains negative and significant in all the different samples.

4.4.3. *Alternative measure of fossil fuel securities*

Panel C presents the results with different measures of fossil fuel securities. In row (12), we exclude Pacific Gas and Electric bonds

[45] These states include California, Connecticut, Delaware, Maine, Maryland, Massachusetts, New Hampshire, New Jersey, New York, Rhode Island, Vermont and Virginia.

Table 10
Effect of mandatory disclosure on investment returns.

Dep. Var =	*Tot Return*		
	Full	*High Divestment*	*Low Divestment*
	(1)	(2)	(3)
Treat × *Post*	−0.00101**	−0.00206***	−0.000214
	(-2.24)	(-2.84)	(-0.28)
Gross Premiums	−0.0000794	−0.0000807	−0.0000765
	(-0.50)	(-0.50)	(-0.46)
Assets	0.000505***	0.000532***	0.000565***
	(2.64)	(2.69)	(3.00)
Investment Income (%)	0.000158	0.000107	0.000178
	(0.66)	(0.47)	(0.70)
Constant	0.0274***	0.0266***	0.0260***
	(11.19)	(10.20)	(10.26)
Wald Test	–	−0.001846	
(p-value)	–	0.068	
Firm FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
N	11,320	9971	9969
R-squared	0.505	0.491	0.490

Notes. This table shows the results of the effect of the California Rule on investment returns. The sample consists of yearly observations. The dependent variable *Tot Return* is the annual return for the firm's bond portfolio. *Treat* × *Post* is a binary variable that equals one for large California licensed insurers after the announcement of the California Rule (i.e. 12/31/2016–2019). In Columns (2) and (3), we examine *Tot Return* and partition the sample with *High Divestment* denoting disclosing firms whose decline in fossil fuel securities from the pre-to the post-period is above the median and *Low* below the median. All columns include firm and year fixed effects. Variable definitions are in Appendix A. Standard errors are clustered at the state-year level. T-stats are displayed in parentheses. *, **, and *** indicate significance levels of 10 %, 5 % and 1 %, respectively.

from the list of fossil fuel securities to ensure that the bankruptcy of the Californian utility in January 2019 is not driving our results. In rows (13) and (14), we classify fossil fuel securities based on their NAICS[46] or SIC[47] industry classification. We consider municipal utilities as electric utilities and classify them as such. The coefficient of interest remains negative and significant across all of these different measures of fossil fuel securities.

5. Conclusion

This paper examines the effect of mandatory disclosure of fossil fuel investments on institutional investors' (insurance companies) portfolio and policy choices. Using a California ruling by the state's insurance department that publicized the fossil fuel holdings of insurers licensed in the state, we find the disclosure results in a significant reduction in fossil fuel holdings. On average, disclosing insurers reduce their holdings by approximately 20 percent relative to non-disclosers. They also become more likely to adopt risk and investment management policies related to climate change. Even after the disclosure ceases, we find the disclosing insurers do not revert to their pre-policy holdings of fossil fuel investments, nor do they change their investment policies once they are in place. We further find the insurers subject to pressure from external stakeholders – both environmental activists and public shareholders are more likely to divest.

It is important to note that the findings are subject to limitations. While our setting helps with identification, it limits the study's generalizability as we focus on a single type of institutional investor and a single disclosure regulation. We call on future research to study other sustainability disclosures impacting other types of institutional investors. Limitations aside, this paper contributes to the

[46] The NAICS industries include: Coal Mining (2121), Oil and Gas Extraction (2111), Support Activities for Mining (2131), Drilling Oil and Gas Wells (213111), Support Activities for Oil and Gas Operations (213112), Support Activities for Coal Mining (213113), Electric Power Generation (22111), Natural Gas Distribution (2212), Electric Power Transmission, Control, and Distribution (22112), Pipeline Transportation of Crude Oil (4861); Pipeline Transportation of Natural Gas (4862), and Other Pipeline Transportation (4869).

[47] The SIC industries include: Bituminous Coal and Lignite Mining (122). Anthracite Mining (123), Coal Mining Services (124), Crude Petroleum and Natural Gas (131), Natural Gas Liquids (132), Oil and Gas Field Services (138), Pipelines, Except Natural Gas (461), Electric Services (491), Gas Production and Distribution (492), and Combination Electric and Gas, and Other Utility (493).

Table 11
Sensitivity tests.

	Coef.	T-Stat
	(1)	(2)
Panel A: Alternative Specifications		
(1) No Firm or Year FEs	−0.009***	(-4.24)
(2) Parent Company FEs	−0.0130***	(-2.63)
(3) Cluster SE by quarter-year	−0.0130***	(-14.49)
(4) Treat × Controls, Firm and Year FEs	−0.0127***	(-5.67)
(5) Post × Controls, Firm and Year FEs	−0.0143***	(-6.18)
(6) One-to-one propensity score matching	−0.0126***	(-4.33)
Panel B: Alternative Samples		
(7) Exclude Californian non-disclosing insurers	−0.0134***	(-5.81)
(8) Restrict to only NAIC climate survey states	−0.0125***	(-4.39)
(9) Restrict to only Carbon Cap-and-Trade states	−0.012***	(-4.64)
(10) Annual instead of quarterly holdings	−0.0143***	(-4.11)
(11) Restrict to above and below $100 million premium threshold	−0.0618***	(-3.51)
Panel C: Alternative Measures of Fossil Fuel Investments		
(12) Excluding Pacific Gas & Electric securities	−0.013***	(-5.68)
(13) NAICS industry classification	−0.0024***	(-3.43)
(14) SIC industry classification	−0.0023***	(-2.66)

Notes: This table shows the sensitivity results. The sample consists of quarterly observations unless otherwise specified. The dependent variable is *Fossil Fuel Securities/Tot Bonds*. *Treat* × *Post* is a binary variable that equals one for large California licensed insurers after the announcement of the California Rule (i.e. 12/31/2016–2019). Row (1) runs the main specification with no firm or year fixed effects. Row (2) substitutes firm fixed effects for parent company fixed effects. Row (3) clusters the standard errors by quarter-year. Rows (4) and (5) control for firm and year fixed effects and the interaction of *Treat* with the controls and *Post* with the controls, respectively. Row (6) uses one-to-one propensity score matching without replacement and matches on gross premiums, total assets, the value of the bond portfolio and the average value of the fossil fuel investments held by the insurer in the pre-period. In row (7) we exclude the Californian insurers below the $100 million premium threshold. In row (8) we restrict the sample to only insurers domiciled in states that are part of the annual NAIC climate survey. In row (9) we restrict the sample to only insurers domiciled in states with carbon cap-and-trade programs. In row (10) we use end of the year portfolio holdings instead of quarterly holdings. In row (11) we restrict the sample to Californian insurers with observations immediately above or below the $100 million premium threshold. In row (12), we exclude Pacific Gas and Electric bonds from the list of fossil fuel securities. Rows (13) and (14) we classify fossil fuel securities based on their NAICS and SIC industry classifications. All rows control for *Gross Premiums, Assets* and *Investment Income (%)* and employ firm and year fixed effects unless otherwise stated. Variable definitions are in Appendix A. Standard errors are clustered at the state-year level. T-stats are displayed in parentheses. *, **, and *** indicate significance levels of 10 %, 5 % and 1 %, respectively.

literature on mandatory non-financial disclosures and the growing literature on environmental disclosures. It also adds to the burgeoning literature on climate finance and specifically answers the call for more research on divestment and stranded assets (Hong et al., 2020). Finally, the paper helps inform regulators who are proposing and adopting California Rule-type policies that require institutional investors to disclose their fossil fuel investments. This paper provides some large-sample evidence that mandatory disclosure does influence investors' portfolios and policy choices.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix D. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jacceco.2025.101829.

Appendix A. Variable Definitions

Variable	Description	Data Source
Assets	Log of insurer's total assets at the end of the fiscal year	Bureau van Dijk Orbis
Assets (1,0)	Indicator variable equal to one if the insurers has total assets greater than the median and zero otherwise.	Bureau van Dijk Orbis
California HQs	Indicator variable equal to one if the insurer has their headquarters in California.	NAIC

(*continued on next page*)

(*continued*)

Variable	Description	Data Source
California Premiums (%)	The percentage of an insurers' premiums that come from customers in California.	CDI website + Bureau van Dijk Orbis
Coal	Indicator variable equal to one for firms in the coal extraction industry	CDI website
Coal Prices	Quarterly change in the price of coal	FRED
Environmental Activists	Indicator variable equal to one if the insurer was the target of an environmental activist campaign from Greenpeace, Unfriend Coal, Insure Our Future or Public Citizen prior to 2016.	Company Websites
Fossil Fuel Securities/Tot Bonds	Par value of fossil fuel investments divided by the total par value of the firm's bond portfolio	NAIC + CDI Website
Fossil Fuel Investments ($M)	Par value of fossil fuel investments	NAIC + CDI website
Gross Premiums	Log of insurer's gross premiums	Bureau van Dijk Orbis
High Divestment	Indicator variable equal to one for disclosing firms who show a decline in fossil fuel securities from the pre- to the post-period that is above the median value.	NAIC + CDI Website
Investment Income (%)	Percentage of insurer's income derived from the investment portfolio	Bureau van Dijk Orbis
Low Divestment	Indicator variable equal to one for disclosing firms who show a decline in fossil fuel securities from the pre- to the post-period that is below the median value.	NAIC + CDI Website
Market Value	Indicator variable equal to one for publicly traded firms with a market valuation above the median and zero otherwise.	Comp-CRSP merged dataset
Natural Gas Prices	Quarterly change in the price of natural gas	FRED
Oil & Gas	Indicator variable equal to one for firms in the oil and gas extraction industries	CDI website
Oil Prices	Quarterly change in the price of oil	FRED
Other	Indicator variable equal to one for fossil fuel securities not in one of those three preceding groups	CDI website
Policy Change	Indicator variable equal to one if the insurer changes their answer (from "No to "Yes") to the question - does the company have a climate change policy with respect to risk management and investment management?" - after the onset of the California Rule	NAIC climate survey
Portfolio	Indicator variable equal to one or zero if the firm replies "yes" or "no" to the following question: "has the company considered the impact of climate change on its investment portfolio?"	NAIC climate survey
Post	Indicator variable equal to one for the period after the announcement of the California Rule (from 12/31/2016 onwards) and zero for the prior period.	CDI website
Public	Indicator variable equal to one if the insurer is publicly-traded.	Bureau van Dijk Orbis
Risk & Investment Mgmt Policy	Indicator variable equal to one or zero if the firm replies "yes" or "no" to the following question: "does the company have a climate change policy with respect to risk management and investment management?"	NAIC climate survey
Sold to Related Insurer	Indicator variable equal to one if the insurer sells a fossil fuel investment to an insurer with the same parent company and zero otherwise. A security is considered sold to another insurer if they buy the security (with same 9-digit CUSIP) within one week of the selling insurer disposing of the security.	NAIC
Strategy	Indicator variable equal to one or zero if the firm replies "yes" or "no" to the following question: "has [the insurer] altered its investment strategy in response to these considerations?"	NAIC climate survey
Tot Return	Annual return for the firm's bond portfolio	NAIC
Total Bond Fair Value ($M)	Total par value of the insurer's bond portfolio	NAIC
Treat	Equal to one for California-licensed insurers and zero otherwise	CDI website
Utilities	Indicator variable equal to one for utilities	CDI website

Appendix B. Screenshot of the California Department of Insurance Website

Below is a screenshot of the California Department of Insurance (CDI) website where information on the insurer's fossil fuel holdings is summarized and displayed to the public. The table includes the insurer's NAIC number, company name, California premiums, assets under management, fossil fuel investments including coal, percentage of fossil fuel investments to assets under management, value of coal investment over the CDI threshold, and value of fossil fuel investments over the CDI threshold.

NAIC	Company ↑	Questionnaire Results	California Total Premium	Assets Under Management	Total Fossil Fuel Investments Including Coal	% of Fossil Fuel to Assets Under Management	Coal Investments Over Threshold	Fossil Fuel Investments Over CDI Threshold (excluding Coal over Threshold)
34789	21ST CENTURY CENTENNIAL INS CO		$2,117,908	$598,355,462	$3,508,995	1%	$0	$0
12963	21ST CENTURY INS CO		$528,510,741	$991,080,094	$48,599,784	5%	$0	$30,275,132
32220	21ST CENTURY NORTH AMERICA INSURANCE COMPANY		$0	$584,428,733	$23,172,511	4%	$7,605,084	$5,549,892
80985	4 EVER LIFE INS CO		$40,715,664	$173,968,567	$9,080,023	5%	$2,141,967	$5,051,194
77879	5 STAR LIFE INS CO		$8,212,749	$237,063,893	$33,342,539	14%	$3,796,338	$19,432,302
71854	AAA LIFE INS CO		$316,146,964	$573,934,317	$86,884,446	15%	$26,888,578	$41,967,785
71471	ABILITY INS CO		$923,656	$1,164,222,449	$58,724,296	5%	$16,300,983	$33,138,996
20010	ACCEPTANCE IND INS CO		$4,256,132	$307,226,780	$23,195,318	8%	$731,500	$15,047,709
10166	ACCIDENT FUND INS CO OF AMER		$14,351,016	$2,865,106,701	$158,003,804	6%	$56,692,236	$73,866,845
62200	ACCORDIA LIFE & ANN CO		$139,546,870	$8,293,793,736	$800,337,343	10%	$194,610,654	$370,836,156
26379	ACCREDITED SURETY & CAS CO INC		$2,657,846	$144,124,279	$2,660,869	2%	$554,049	$1,546,454
22667	ACE AMER INS CO		$825,588,123	$9,202,767,375	$545,994,692	6%	$125,165,647	$266,919,671
20702	ACE FIRE UNDERWRITERS INS CO		$11,799,944	$99,262,882	$2,474,139	2%	$433,428	$1,436,406
20699	ACE PROP & CAS INS CO		$89,716,070	$6,208,044,589	$287,160,283	5%	$60,835,953	$108,455,654
19984	ACIG INS CO		$3,980,110	$453,140,343	$7,840,596	2%	$763,205	$3,786,575
22950	ACSTAR INS CO		$34,761	$51,763,922	$4,791,873	9%	$2,214,825	$1,334,419

Appendix C. Loya Insurance Company's Response to the NAIC Survey (2018)

Question (1): Does the company have a climate change policy with respect to risk management and investment management?"

Response: No. The Company's lines of business are private automobile policies in the nonstandard sector, with auto liability and auto physical damage accounting for approximately 65 % and 35 % of total premiums written, respectively. As such, the Company does not believe climate change will have significant impact in its operations with regards to risk management, other than catastrophic losses described in question #3 below. With regards to investment management, the Company has not modified its existing investment policy as it pertains to climate change. However, the Company will take into consideration California's Climate Risk Carbon Initiative and refrain from making any new investments in thermal coal and/or fossil fuel.

Question (2a): Has the company considered the impact of climate change on its investment portfolio?

Question (2b): Has [the insurer] altered its investment strategy in response to these considerations?

Response: Yes. No. The Company's investment policy is to acquire and hold NAIC Grade Class 1 and 2 securities. The securities held by the Company consist primarily of municipal/state and corporate bonds from a diversified number of issuers. As such, the Company does not believe it has significant exposure to climate change risks with regards to its investment holdings. However, the Company will take into consideration California's Climate Risk Carbon Initiative request and refrain from making any new investments in thermal coal and/or fossil fuel.

References

Abramova, I., Core, J., Sutherland, A., 2020. Institutional investor attention and firm disclosure. Account. Rev. 95, 1–21.
Agarwal, V., Gay, G., Ling, L., 2014. Window dressing in mutual funds. Rev. Financ. Stud. 27, 3133–3170.
Al Khaffaf, I., Tamimi, A., Ahmed, V., 2024. Pathways to carbon neutrality: a review of strategies and technologies across sectors. Energies 17 (23), 6129.
Allen, L., Saunders, A., 1992. Bank window dressing: theory and evidence. J. Bank. Finance 16, 585–623.
Anantharaman, D., 2017. The role of specialists in financial reporting: evidence from pension accounting. Rev. Account. Stud. 22, 1261–1306.
Ansar, A., Caldecott, B., Tilbury, J., 2013. Stranded Assets and the Fossil Fuel Divestment Campaign: what does Divestment Mean for the Valuation of Fossil Fuel Assets? *Stranded Assets Programme*, pp. 1–81.
Bank of England, 2015. The Impact of Climate Change on the UK Insurance Sector, pp. 1–87.
Barber, B., Morse, A., Yasuda, A., 2021. Impact investing. J. Financ. Econ. 139, 162–185.
Bird, A., Karolyi, S., 2016. Do institutional investors demand public disclosure? Rev. Financ. Stud. 29, 3245–3277.
Bloomberg, 2016. California urges insurers to divest coal bets to cut risk. Bloomberg. Available at https://www.bloomberg.com/news/articles/2016-01-25/california-urges-insurers-to-divest-from-coal-bets-to-cut-risk#xj4y7vzkg.
Breuer, M., deHaan, E., 2024. Using and interpreting fixed effects models. J. Account. Res. 62, 1183–1226.
Broccardo, E., Hart, O., Zingales, L., 2022. Exit versus voice. J. Polit. Econ. 130, 3101–3145.
Brownen-Trinh, R., Orujov, A., 2023. Corporate socio-political activism and retail investors: evidence from the black lives matter campaign. J. Corp. Finance 80, 102417.
Cahan, S., 1992. The effect of antitrust investigations on discretionary accruals: a refined test of the political-cost hypothesis. Account. Rev. 67, 77–95.
California Department of Insurance, 2022. Insurers overview. http://www.insurance.ca.gov/0250-insurers/#:~:text=The%20Insurance%20market%20in%20California,maintain%20an%20economically%20healthy%20marketplace.
Carlin, D., 2021. The case for fossil fuel divestment. Forbes. Available at https://www.forbes.com/sites/davidcarlin/2021/02/20/the-case-for-fossil-fuel-divestment/?sh=57c2472076d2.
Ceccarelli, M., Glossner, S., Homanen, M., 2021. Catering Through Transparency: Voluntary ESG Disclosure by Asset Managers and Fund Flows. Working paper.
Ceccarelli, M., Ramelli, S., Wagner, A., 2022. Low-Carbon Mutual Funds. Working paper.
Chen, T., Dong, H., Lin, C., 2020. Institutional shareholders and corporate social responsibility. J. Financ. Econ. 135, 483–504.
Choi, D., Gao, Z., Jiang, W., Zhang, H., 2021. Global Carbon Divestment and Firms' Actions. Working Paper.
Christensen, H., De George, E., Joffre, A., Macciocchi, D., 2023. Consumer Response to the Revelation of Corporate Social Irresponsibility. Working Paper.
Cohen, L., Rouen, E., Sachdeva, K., 2025. Widespread Misstatements of Emissions by Major U.S. Corporations. Working Paper.
Dai, R., Liang, H., Ng, L., 2021. Socially responsible corporate customers. J. Financ. Econ. 142, 598–626.
Dai, J., Ormazabal, G., Penalva, F., Raney, E., 2024. Imposing Disclosure on Investors that Choose to Promote Sustainability: Does it Lead to Portfolio Decarbonization? Working Paper.
Darnall, N., Henriques, I., Sadorsky, P., 2010. Adopting proactive environmental strategy: the influence of stakeholders and firm size. J. Manag. Stud. 47, 1072–1094.
Dentons, 2016. California commissioner calls for divestiture from carbon-based investments. https://www.dentons.com/en/insights/newsletters/2016/february/5/insurance-regulation-update/ireg-update.
Dewatripont, M., Jewitt, I., Tirole, J., 1999. The economics of career concerns, part I: comparing information structures. Rev. Econ. Stud. 66, 183–198.
Doak, J., Ridling, J., Roberston, S., Atkins, N., Rosendale, Jr.M., Godfread, J., 2016. Open letter to Dave Jones. https://gray-arc-content.s3.amazonaws.com/KFYR/Multi-State%20Letter%20to%20Commissioner%20Dave%20Jones.pdf.
Downar, B., Ernstberger, J., Reichelstein, S., Schwenen, S., Zaklan, A., 2021. The impact of carbon disclosure mandates on emissions and financial operating performance. Rev. Account. Stud. 26, 1137–1175.
Duran, P., 2021. New Zealand Passes Climate Change Disclosure Laws for Financial Firms in World First. Reuters. Available at https://www.reuters.com/business/sustainable-business/new-zealand-passes-climate-change-disclosure-laws-financial-firms-world-first-2021-10-21/.
Dyck, A., Lins, K., Roth, L., Wagner, H., 2019. Do institutional investors drive corporate social responsibility? International evidence. J. Financ. Econ. 131, 693–714.
ERM, 2022. Cost of climate disclosure survey. https://www.sustainability.com/globalassets/sustainability.com/thinking/pdfs/2022/climate-disclosure-survey_fact-sheet-april-2022.pdf.
Europe Securities and Markets Authority, 2022. ESAs propose disclosures for fossil gas and nuclear energy investments. https://www.esma.europa.eu/press-news/esma-news/esas-propose-disclosures-fossil-gas-and-nuclear-energy-investments.
European Parliament, 2024. New social and environmental reporting rules for large companies. https://www.europarl.europa.eu/news/en/press-room/20220620IPR33413/new-social-and-environmental-reporting-rules-for-large-companies.
Evain, J., Cardona, M., Nicol, M., 2018. Article 173: Overview of climate-related Financial Dislosure After Two Years of Implementation. Working paper.
Fineman, S., Clarke, K., 1996. Green stakeholders: industry interpretations and reponse. J. Manag. Stud. 33, 715–730.
Gehricke, S., Leippold, M., Schimanski, T., 2025. To Disclose, or Not to Disclose: Evaluating the Effectiveness of Mandatory Climate-Related Disclosure. Working Paper.
Goldman Sachs, 2023. Investment policy statements | an overview for board members. https://privatewealth.goldmansachs.com/public/ips-white-paper-2023.pdf.
Gormley, T., Matsa, D., 2014. Common errors: how to (and not to) control for unobserved heterogeneity. Rev. Financ. Stud. 27, 617–661.
Greenhut, S., 2016. Coal Divestment and the California Insurance Industry. R Street Institute, pp. 1–11.
Greenwald, J., 2010. Poizner fights to ban insurers from investing in Iran. Bus. Insur. Available at https://www.businessinsurance.com/article/00010101/STORY/311149979/Poizner-fights-to-ban-insurers-from-investing-in-Iran.
Grundl, H., Dong, M., Gal, J., 2016. The evolution of insurer portfolio investment strategies for long-term investing. OECD Journal: Financ. Mark. Trends 1–55.
Hainmueller, J., 2012. Entropy balancing for causal effects: a multivariate reweighting method to produce balanced samples in observational studies. Polit. Anal. 20, 25–46.
Halper, J., Bussiere, S., Shriver, T., 2022. Asset management industry confronts the challenges presented by climate change transition. Harvard Law School Forum on Corporate Governance. Available at https://corpgov.law.harvard.edu/2022/02/28/asset-management-industry-confronts-the-challenges-presented-by-climate-change-transition/.
Hombach, K., Sellhorn, T., 2019. Firm Value Effects of Targeted Disclosure Regulation: the Role of Reputational Costs. Working Paper.
Hong, H., Karolyi, G.A., Scheinkman, J., 2020. Climate finance. Rev. Financ. Stud. 33, 1011–1023.
Humphrey, J., Li, Y., 2021. Who goes green: reducing mutual fund emissions and its consequences. J. Bank. Finance 126, 1–17.
Hyperstat, 2023. Ca.gov - insurance.ca.gov. https://hypestat.com/info/insurance.ca.gov.
Ilhan, E., Krueger, P., Sautner, Z., Starks, L., 2021. Climate Risk Disclosure and Institutional Investors. Working paper.
Insurance Information Institute, 2021. 2021 insurance fact book. https://www.iii.org/sites/default/files/docs/pdf/insurance_factbook_2021.pdf.
Insurance Journal, 2018. $332B in premiums makes California world's 4th largest insurance market. https://www.insurancejournal.com/news/west/2018/04/05/485527.htm.
Jones, D., 2016a. Coal divestment & carbon-based investment data call. California Depart. Insuran. Available at http://www.insurance.ca.gov/0400-news/0100-press-releases/archives/statement010-16.cfm.
Jones, D., 2016b. Coal divestment & carbon-based investment data call - clarification. California Depart. Insuran. Available at https://www.insurance.ca.gov/0250-insurers/0300-insurers/0100-applications/ci/upload/Coal-Divestment-Carbon-Data-Call-Clarification-Letter-Large-Companies-v8.pdf.
Jouvenot, V., Krueger, P., 2019. Mandatory Corporate Carbon Disclosure: Evidence from a Natural Experiment. Working paper.
Krueger, P., Sautner, Z., Starks, L., 2019. The Importance of Climate Risks for Institutional Investors. Working paper.

Lehman, R., 2016. Jones' coal divestment call is irresponsible, blatantly political. Roads Streets. Available at https://www.insurancejournal.com/blogs/right-street/2016/01/25/396284.htm.

Lester, R., Olbert, M., 2024. Firms' Real and Reporting Responses to Taxation: a Review. Working paper.

Lewis, R., O'Donovan, G., Willett, R., 2017. The effect of environmental activism on the long-run market value of a company: a case study. J. Bus. Ethics 140, 455–476.

Messervy, M., 2016. Insurer Climate Risk Disclosure Survey Report & Scorecard: 2016 Findings & Recommendations. Ceres, pp. 1–80.

Messonier, J., Nguyen, B., 2021. Showing off Cleaner Hands: Mandatory climate-related Disclosure by Financial Institutions and the Financing of Fossil Energy. Working paper.

NAIC, 2021. U.S. insurance industry's cash and invested assets surpass $8 trillion at year-end 2021. https://content.naic.org/sites/default/files/capital-markets-special-reports-asset-mix-ye2021.pdf.

NAIC, 2022. Growth in U.S. insurance industry's cash and invested assets declines to 1.3% at year-end 2022. https://content.naic.org/sites/default/files/capital-markets-special-reports-asset-mix-ye2022.pdf.

NAIC, 2023. NAIC climate risk disclosure survey. http://www.insurance.ca.gov/0250-insurers/0300-insurers/0100-applications/ClimateSurvey/.

Rabinowitz, D., 2019. Investments by insurers: how they're regulated and why it matters. N. Y. Law J. 262, 1–3.

Reitmeyer, J., 2022. Divesting from fossil fuels gets first look. NJ Spotlight News. Available at https://www.njspotlightnews.org/2022/08/fossil-fuels-clean-energy-divestment-public-worker-pension-funds-senate-environment-and-energy-committee-chair-bob-smith-climate-change-sea-level-rise-global-warming-divestnj-carbon-content-esg-policy/.

Reuters, 2016. California insurance commissioner calls for coal divestment. Reuters. Available at https://www.reuters.com/article/california-insurance-coal/california-insurance-commissioner-calls-for-coal-divestment-idINL2N15921L.

Ring, E., 2024. Climate Action' Has California's Energy Economy on its Knees. WSJ. Available at https://www.wsj.com/opinion/climate-action-has-californias-energy-economy-on-its-knees-green-lawfare-oil-e11518a2?mod=hp_opin_pos_3#cxrecs_s.

Schroders, 2023. Institutional investor study 2023. https://www.schroders.com/en-us/us/institutional/institutional-investor-study-2023/.

Securities and Exchange Commission, 2022. Enhanced disclosures by certain investment advisers and investment companies about environmental, social, and governance investment practices. https://www.sec.gov/rules/proposed/2022/ia-6034.pdf.

Teu, V., 2021. Management and disclosure of climate-related risks by Hong Kong fund managers. Dentons. Available at https://hongkong.dentons.com/en/insights/alerts/2021/november/8/management-and-disclosure-of-climate-related-risks.

Tomar, S., 2023. Greenhouse gas disclosure and emissions benchmarking. J. Account. Res. 61 (2), 451–492.

Treacy, J., Sako, Y., Yeu, S., 2022. Esg regulatory developments in the UK, Japan, and Hong Kong. K&L Gates Hub. Available at https://www.klgates.com/ESG-Regulatory-Developments-in-the-UK-Japan-and-Hong-Kong-1-14-2022.

Tsang, A., Xie, F., Xin, X., 2019. Foreign institutional investors and corporate voluntary disclosure around the world. Account. Rev. 94, 319–438.

UNFCC: The United Nations Framework Convention on Climate Change, 2022. What is climate finance? https://www.climateinvestmentchallenge.org/about-climate-finance.

Vetter, D., 2020. JPMorgan chase tops dirty list of 35 fossil fuel-funding banks. Forbes. Available at https://www.forbes.com/sites/davidrvetter/2020/03/18/jpmorgan-chase-tops-dirty-list-of-35-fossil-fuel-funding-banks/?sh=519c0d875dfe.

Weil, D., Fung, A., Graham, M., Fagotto, E., 2006. The effectiveness of regulatory disclosure policies. J. Pol. Anal. Manag. 25, 155–181.